



Does Starting Universal Childcare Earlier Influence Children's Skill Development

Author(s): Daniel Kuehnle and Michael Oberfichtner

Source: *Demography*, February 2020, Vol. 57, No. 1 (February 2020), pp. 61-98

Published by: Duke University Press on behalf of the Population Association of America

Stable URL: <https://www.jstor.org/stable/10.2307/48681402>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



Population Association of America and Duke University Press are collaborating with JSTOR to digitize, preserve and extend access to *Demography*

JSTOR

Does Starting Universal Childcare Earlier Influence Children's Skill Development?

Daniel Kuehnle^{1,2} · Michael Oberfichtner³

Published online: 15 January 2020
© Population Association of America 2020

Abstract

As many developed countries enact policies that allow children to begin universal childcare earlier, understanding how starting universal childcare earlier affects children's cognitive and noncognitive skills is an important policy question. We provide comprehensive evidence on the multidimensional short- and longer-run effects of starting universal childcare earlier using a fuzzy discontinuity in the age at starting childcare in Germany. Combining rich survey and administrative data, we follow one cohort from age 6 to 15 and examine standardized cognitive test scores, noncognitive skill measures, and school track choice in a unified framework. Children who start universal childcare four months earlier around age 3 do not perform differently in terms of standardized cognitive test scores, measures of noncognitive skills, school track choice, or school entrance examinations. We also find no evidence of skill improvements for children with low socioeconomic status, although we provide suggestive evidence that they may benefit from high-quality care. Our estimates refer to children who start childcare before they become legally entitled, for whom the literature would predict low gains to starting childcare earlier. We provide further evidence on this relationship between parental resistance to and children's potential gains from childcare. Simply allowing children to start universal childcare earlier is hence not sufficient to improve children's skill development, particularly for children with low socioeconomic status.

Keywords Universal childcare · Child development · Skill formation · Cognitive skills · Noncognitive skills

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s13524-019-00836-9>) contains supplementary material, which is available to authorized users.

✉ Daniel Kuehnle
daniel.kuehnle@uni-due.de

¹ Universität Duisburg-Essen, Weststadttürme Berliner Platz 6-8, 45127 Essen, Germany

² IZA Institute of Labor Economics, Bonn, Germany

³ Institute for Employment Research (IAB), Regensburger Str. 100, 90478 Nuremberg, Germany

Introduction and Motivation

Over the last decades, many countries have undergone profound changes in their demographic, social, and economic situations: across OECD countries, fertility rates have declined (OECD 2016a), and maternal employment rates have increased (OECD 2016b). These two stylized facts are inherently linked through the provision of early childhood education and care (ECEC). Indeed, policymakers, parents, and social activists in many countries still intensively discuss the resulting opportunities and challenges. Proponents of public childcare point out that it may increase maternal employment rates (see, e.g., Baker et al. 2008; Nollenberger and Rodríguez-Planas 2015) and fertility rates (see, e.g., Bauernschuster et al. 2016). Thus, from a demographic perspective, providing ECEC takes a central role as a promising policy response to the challenge of shrinking working-age populations in aging societies.

More controversial are the effects of attending public childcare on children's development, which is crucial for sustainable long-run development. Among others, Lutz (2014) therefore has argued that policymakers should focus on strengthening the human capital base, beginning in early childhood. Many critics of expanding ECEC programs, however, worry that attending public (or other forms of external) childcare may negatively impact children's development and deteriorate the human resource base of the future (see discussions in Brooks-Gunn et al. 2002; Hsin and Felfe 2014). In contrast, proponents argue that attending public childcare improves children's development and strengthens the human resource base of the future (e.g., Nores and Barnett 2010). Understanding how attending childcare affects children's development—and thus countries' future human capital—is hence important from a demographic perspective (on the importance of human capital trends more broadly for demographic changes, see Lutz et al. 2014).

Consistent with the optimistic view, access to universal childcare programs has greatly expanded across industrialized countries since the 1970s (OECD 2011), and most children now participate in some form of ECEC: by age 5, almost all children in OECD countries attend formal childcare, with an average attendance rate of 96% as of 2014 among countries reporting enrollment rates for all age groups (OECD 2017). Enrollment rates for younger children, however, are substantially lower, at 38% for children below age 3. Given that almost all children attend childcare the year before school entry, the main policy question no longer pertains to the extensive margin of attendance but rather to policies that allow children to start childcare earlier—that is, to the intensive margin. One recent example is the “3-K for All” initiative in New York City, which aims to entitle all children to start childcare at age 3 instead of age 4 (Taylor 2017).

To justify expanding universal ECEC programs, policymakers often refer to the positive effects of programs targeting disadvantaged children on their later-life outcomes (e.g., Heckman et al. 2013).¹ However, the evidence on the effects of universal childcare programs on children's outcomes reveals rather inconclusive results, with some studies finding positive average effects in the short run (e.g., Drange and Havnes

¹ The literature distinguishes between targeted programs that focus only on certain (disadvantaged) groups and universal programs that do not base eligibility on a measure of disadvantage. See Bernal and Keane (2011), Ruhm and Waldfogel (2012), and Elango et al. (2016) for excellent reviews of the prior literature.

2019) and others documenting no effects (e.g., Carta and Rizzica 2018) or even negative effects (e.g., Baker et al. 2008). Further, evidence covering manifold outcomes in the short and long run is available only for the vast expansion of universal childcare in Quebec (Baker et al. 2008, 2019). The sobering results from Quebec, though, do not necessarily generalize to today's ECEC programs for at least two reasons. First, Quebec's childcare program is often considered to be of low quality (e.g., Haeck et al. 2015), although Baker et al. (2019) argued that the quality of care was similar to that of other developed countries at that time. Second, the expansion of childcare both reduced the age at which children started childcare and drew additional children into childcare. To inform current policy debates, we thus still lack comprehensive evidence on the short- and longer-run effects of starting specific types of ECEC programs earlier on broad sets of outcomes.

To address this limitation, we investigate the effects of starting childcare earlier on a comprehensive set of skill measures at ages 6 and 15. Our study makes two contributions to the literature. First, we examine short- and longer-run effects of starting universal childcare earlier in a unified framework. Neuroscience identified early childhood as a critical period for brain development (Shonkoff and Phillips 2000), implying that early investments foster later skill developments, referred to as "dynamic complementarities" (Cunha et al. 2010). Given such complementarities, disadvantaged children could benefit permanently from early human capital investments, such as starting childcare earlier. Because children in our treatment and control groups start and attend school together, they are affected by the same school environment, and we can thus examine short- and longer-run outcomes. To this end, we combine administrative and survey data to follow one birth cohort of children over time.

Our second contribution lies in providing comprehensive evidence within a unified framework for the effect of starting universal childcare earlier on different skill dimensions. Our outcome variables comprise standardized cognitive test scores (e.g., language, math, and general cognition), noncognitive/socioemotional skill measures (e.g., Big Five personality traits, Strength and Difficulties Questionnaire) as well as school readiness and school track choice as broader measures of ability. The different skill dimensions strongly affect later educational attainment, labor market outcomes, and health (Almlund et al. 2011). This multidimensionality of outcomes is important. For example, Havnes and Mogstad (2011) found no effect of universal childcare on test scores but did find an effect on educational attainment and earnings. Their results suggest that improvements in noncognitive skills, rather than cognitive skills, drive the positive effect of ECEC on education and earnings.

To estimate the causal effect of starting formal childcare earlier, we exploit exogenous variation in the age at which children start childcare within a fuzzy regression discontinuity design. Although children in our sample are legally entitled to a slot in public childcare from their third birthday onward, some children start formal childcare earlier. Typically, children start childcare in the summer of the calendar year in which they turn 3. Therefore, many children born toward the end of a year start childcare before their third birthday; in contrast, children born at the beginning of the subsequent calendar year start childcare after their third birthday. This enrollment pattern creates a December/January five-month discontinuity in the childcare starting age. Importantly, our data show that children on the two sides of the threshold do not differ in relevant observable characteristics, such as parents' native language, age, and education.

Furthermore, we control for age at testing, which thus cannot confound our estimated treatment effect.

The estimated treatment effect stems from three channels. First, children born toward the end of the year, on average, start childcare at a lower absolute age. Second, they also stay longer in childcare because the December/January threshold does not affect children's school entrance. Third, children starting before age 3 will likely be the youngest children when starting childcare. The estimated effect hence combines starting childcare at a lower absolute and relative age and attending childcare longer, resembling parents' decision about when to enroll their children.²

We consistently find no effect of starting childcare earlier across three data sets. At age 15, children who start childcare earlier do not differ regarding cognitive test scores, noncognitive skills, and school track choice. The estimates are precise enough to rule out that starting universal childcare earlier substantially affects skills, on average. Next, we analyze the short-run effect on children's skills measured during school entrance examinations around age 6 to analyze potential effect fade-out. The data from school entrance examinations also remedy concerns that any effects of absolute or relative age at school entry might contaminate our results because school entry examinations take place before any such effects can evolve. Examining children's school readiness, language competencies, motor skills, and behavioral problems, we do not find a robust effect for our complete sample.

The effects of starting childcare earlier ultimately depend on the difference between the quality of care in the home environment and that in childcare. The substantial effect heterogeneities across socioeconomic subgroups found in the literature are accordingly often attributed to differences in the home care environment as proxied by families' socioeconomic status (SES; see the next section for more details). To account for differences in home care quality, we also investigate the effect for children of nonnative mothers or low-educated mothers. Somewhat surprisingly, we also find no evidence of skill improvements for any subgroup.

The absence of an effect on children's skill development likely reflects that the difference between the quality of care in the home environment and that in childcare are too minor to matter. To probe this explanation, we increase the variation in this difference by examining additional subgroups defined by the quality of childcare offered locally. We expect the largest gains for children from relatively low-quality home environments who enter high-quality childcare programs. This part of our analysis allows only for a tentative interpretation because parents living in districts offering different care quality are likely to differ systematically, given that our measures of childcare quality are rather crude and our sample sizes become relatively small in subgroup analyses. Nevertheless, our results suggest that low-SES children benefit in terms of school readiness from starting high-quality childcare earlier. This pattern is consistent with the positive effects of targeted programs (e.g., Blau and Currie 2006) and the importance of comparing the quality of both home care and formal childcare (e.g., Cascio 2017; Elango et al. 2016). Our findings underpin the importance of

² In contrast, a policy changing all children's enrollment preserves children's relative age. Isolating relative age effects in early childcare hence informs an additional policy question. Although doing so is not possible in our setting, Cascio and Schanzenbach (2016) investigate effects of relative age at kindergarten entry around age 5. They found that being relatively younger did not have a negative effect and that children appear to benefit from having older classmates.

considering not only heterogeneities by SES but also the interaction between the quality of care in the home environment and that in childcare.

Regarding parental resistance against early childcare, Cornelissen et al. (2018) showed that children of parents with a high resistance benefit from starting earlier. Moreover, their point estimates even indicate that starting childcare early might reduce school readiness for children of parents with low resistance. Our treatment effect is estimated for children who start childcare before they become legally entitled. Thus, they plausibly have a low resistance against early childcare, which would predict low gains for them or even negative effects. By analyzing the effect of starting childcare earlier for this group, we provide further evidence on the relationship between parental resistance to and children's potential gains from childcare.

Literature Review

ECEC programs differ in at least three dimensions that likely explain different results between studies: First, some programs target children aged 3 to 6, whereas others target younger children. Second, the structure and quality of programs varies. Third, the alternative to the studied childcare programs range from other public programs (e.g., evaluations of pre-K against Head Start) to informal external care to almost exclusively maternal care. In this section, we review evidence on universal childcare programs for children up to age 5 along these dimensions. To keep the discussion within manageable bounds, we focus on programs similar to German childcare, which is childcare-oriented (as opposed to preschool-oriented) and for which the main alternative care arrangement is maternal care or some other type of informal care.³ Given that the features of childcare programs are relatively homogeneous within countries, we present the evidence by countries before drawing an overall picture.

One well-studied program comes from Quebec, which introduced highly subsidized universal childcare in 1997. The program gradually covered all children aged 0–4, offering full-time center-based childcare. The reform raised childcare attendance by 15 percentage points, and this increase was almost completely driven by children attending for more than 20 hours per week (Baker et al. 2008). Because low-income families had already been eligible for the subsidy, the reform mainly affected middle- and high-income families (Baker et al. 2008; Lefebvre and Merrigan 2008). Assessing children's short-run outcomes up to age 3, Baker et al. (2008) provided evidence for negative reduced-form effects on noncognitive outcomes (−0.1 standard deviations (SD) for motor skills and social development scores) and insignificant effects of similar size for cognitive outcomes. Exploring effect heterogeneities, Kottelenberg and Lehrer (2014)

³ The distinction follows Cascio's (2015) framework to classify ECEC programs. Cascio first differentiated whether a universal program follows a preschool or a childcare orientation: preschool-oriented programs are typically delivered through the public school system with an explicit focus on promoting school readiness, and childcare-oriented programs focus more on providing childcare than on fostering skill development. Second, Cascio also distinguished whether children in the alternative care environment are mainly looked after by their mothers (or receive other types of informal care) or whether children attend another form of center-based childcare. For a comprehensive perspective on both preschool- and childcare-oriented programs, Table A1 in the online appendix concisely summarizes previous studies on the effects of universal childcare on children's skills, focusing on studies that exploit some form of exogenous variation in universal childcare attendance.

showed that children who gained access to childcare at an earlier age experienced more negative effects, and Kottelenberg and Lehrer (2017) showed that the most disadvantaged and low-ability children even benefited from the policy. Recent studies examined the long-run effects of the policy with ambiguous findings. Baker et al. (2019) found that children's health, life satisfaction, and crime rates are negatively affected at age 20, whereas Haeck et al. (2018) did not find persistent negative effects on health and behavior at ages 12–19. Overall, the negative average effects are consistent with the stronger take-up by families who likely provide higher-quality home care and concerns about the initially low quality of the program (e.g., Haeck et al. 2015). However, Baker et al. (2019) found some evidence that the quality of the program was comparable to those of other developed countries at that time.

More evidence for childcare-oriented programs exists for European countries. From 1975 onward, Norway increased the supply of subsidized universal childcare slots for children aged 3–6. To receive federal funds, municipalities had to comply with high federal quality standards. Exploiting variation across time and municipalities, Havnes and Mogstad (2011) found that the program had no effect on children's test scores but reduced their later-life (at age 30) welfare dependency and increased educational attainment and earnings. Havnes and Mogstad (2015) additionally showed that the positive effects are driven by children of low-income parents and that children of upper-income parents experienced negative effects. Drange and Havnes (2019) provided additional evidence for Norway for children born between 2004 and 2006. Exploiting an assignment lottery to estimate the effect of starting childcare at 15 months compared with 19 months, they found that children who started childcare earlier scored higher on standardized language (+0.16 SD) and math tests (+0.1 SD), with substantially larger effects for children of low-educated parents. They argued that differences in childcare starting age, and not in childcare quality or family income, explain the results.

For Denmark, Datta Gupta and Simonsen (2010) used an instrumental variables approach to examine the effects of universal childcare for 3- to 6-year-olds in comparison with home care on noncognitive skills at age 7. Using survey data for children born in 1995, they found no effects of center-based care compared with home care, but they found negative effects of family day care compared with home care for boys with low-educated mothers. Their findings are not entirely consistent with the studies from Norway and Canada, but the studies differ with respect to the quality of the programs and the identifying variation.⁴

For Germany, Cornelissen et al. (2018) exploited the differential rates of expansion of childcare centers across municipalities over time as an instrumental variable within a marginal treatment effects (MTE) approach. They found that attending half-day childcare for (at least) three years between ages 3 and 6 positively affects the school readiness of children who are least likely to enroll early. Felfe and Zierow (2018) exploited a reform increasing the number of full-day slots to investigate whether attending full-day instead of half-day childcare affects children outcomes at age 6. They found an increase in socioemotional problems (by 0.18 SD), particularly for

⁴ Datta Gupta and Simonsen (2016) examined the effect of attending center-based care compared with family day care at age 2 and found positive effects on children's grade point average in Danish language (+0.2 SD) at age 16. The effects are again larger for children from low-educated mothers and point to the importance of quality differences between the different care environments.

disadvantaged children. Felfe and Lalive (2018) examined the effects of starting childcare before age 3 within an MTE approach, again exploiting regional differences in expansion. They found that children most likely to attend childcare experience improvements in motor skill development and that children least likely to attend childcare improve their socioemotional skills. These improvements are stronger for boys and children from families with low education or migration backgrounds.

Additional evidence on the effects for children younger than 3 exists for Italy. Carta and Rizzica (2018) examined how early access to full-day subsidized childcare for 2-year-old children affects maternal employment and children's outcomes. Using an age-eligibility cutoff rule within a difference-in-discontinuities design, they found no effects on children's language and mathematics test scores at age 7. Similar to the reform in Quebec, the Italian reform mostly changed the price of childcare for medium- to high-income families. Ichino et al. (forthcoming) also examined the effect of attending childcare (both half- and full-day) before age 3 for relatively wealthy families in the city of Bologna. Using institutional admission rules within a regression discontinuity design, the study revealed that one additional month of childcare reduces IQ by about 0.045 SD between ages 8 and 14, with the strongest effects for the most-affluent families. The study also provided evidence for similar negative effects on noncognitive skills. The study underlines the importance of qualitatively high one-to-one interactions early in life given that the quality of care provided even in a high-quality public daycare system may be lower than that provided at home by affluent families.

Taken together, the previous literature shows that the quality of childcare programs and the quality of the alternative care mode are crucial to understand the effects of childcare programs. When program quality is held constant, the effects of attending a program become less favorable or even harmful as the quality of the care from an alternative care environment improves. When the quality of the alternative care mode is held constant, the effects of attending a program become more favorable as program quality improves. Most prior studies, however, focused on relatively short-run outcomes, and evidence on longer-run effects is available only for Quebec (e.g., Baker et al. 2019; Haeck et al. 2018), where care was initially of untypically low quality. We thus lack evidence for the longer-run effects of starting other specific childcare programs on broad sets of outcomes.

Public Childcare and Alternative Care Modes in Germany

In this section, we describe the institutional features of the West German childcare system in the late 1990s, when our sample entered childcare.⁵ Childcare is typically organized and funded at the local level, with municipalities (*Gemeinde*) bearing the primary responsibility (Evers and Sachße 2002). The childcare centers are mainly run either by municipalities or by large nongovernmental organizations that cooperate closely with the municipal decision-makers (Heinze et al. 1997). Unlike in the United States or the United Kingdom, for example, no noteworthy private childcare market ever emerged in Germany, mainly because of strict regulations, high market entry

⁵ We focus on West Germany because of the large differences in early childcare attendance between East and West Germany (Kreyenfeld et al. 2001).

barriers, and dominance by publicly funded providers (Kreyenfeld and Hank 2000).⁶ Childcare provision was highly subsidized, so parents typically did not pay more than 3% to 4% of annual household income on childcare services (Evers et al. 2005); for families on social assistance, childcare provision was free.

Although childcare was almost nonexistent for children under age 3,⁷ children from age 3 until school entry became entitled to attend half-day public childcare (*Kindergarten*) in August 1996. The decentralized planning and funding process gave rise to considerable heterogeneity in childcare availability across districts (Mamier et al. 2002). Using official childcare statistics, we examine the distribution of available slots per child aged 3–6 across German districts in 1994, 1998, and 2002 (see Fig. A2, online appendix). In 1994, 50% of children lived in a district where the ratio of childcare slots per child in the eligible age group was 0.86 or higher. Anecdotal evidence suggests that 0.9 slots per child constituted sufficient supply, if not oversupply, which is also in line with the current enrollment rates.⁸ By 1998, the median child lived in a district offering more than one slot per child, and more than 75% of children lived in a district supplying at least 0.9 slots per child. Hence, the decentralized planning process led to an oversupply of childcare slots in many German regions, allowing children to start childcare before becoming legally entitled.

In the late 1990s, children usually spent four hours in the morning in formal childcare and the remaining time in parental or other informal care (Hank and Kreyenfeld 2003). Childcare centers throughout Germany pursued developmental goals for language, physical, and behavioral skills. Because of political decentralization, state governments determine educational policies in Germany. Hence, each state has a separate law regulating the quality and content of childcare. Although the wording differs between the states, the goals are similar.⁹ The educational content follows the social pedagogy tradition in which children develop skills mainly through play and informal learning (Scheiwe and Willekens 2009). This pedagogical approach contrasts with programs that have a stronger school orientation emphasizing formalized forms of early learning (Chartier and Geneix 2006). The average expenditure per attending child was \$4,937 in 1999, which was higher than the OECD average (\$3,847) and the expenditures in France (\$3,901) but lower than in the United Kingdom (\$6,233) and in the United States (\$6,692; see OECD 2002).

When interpreting the estimated effect of starting childcare earlier, we implicitly compare it with a counterfactual care mode. According to the German Socio-Economic Panel (SOEP; Wagner et al. 2007) for West Germany in 1997, only 1.7% of children aged 2–4 were looked after by childcare providers, and 2.1% regularly used paid private care arrangements (e.g., nannies). Instead, 30.1% were regularly looked after by family and relatives. Thus, the counterfactual care mode against which we estimate

⁶ For more information on the administration and funding of the German childcare system, see the descriptions by Kreyenfeld et al. (2001) and Evers and Sachße (2002).

⁷ In 1998, only 1.9 childcare slots, on average, were available for 100 children below the age of 3, and the supply of such slots began to increase only after 2005 because of political reforms. In 2013, children became legally entitled to a slot from their first birthday onward.

⁸ As a case in point, the city of Offenbach offered 0.84 slots per child in 1998 and had trouble filling these slots (Frankfurter Allgemeine Zeitung (FAZ) 1998).

⁹ The legal texts are available at <https://bage.de/service/links-zu-den-kita-gesetzen-der-einzelnen-bundeslaender/>.

the effect of starting childcare earlier is home care, mainly provided by mothers. For children aged 0–2, nonmaternal care is even less prevalent because these children are less likely to use private care arrangements, with only 25% looked after by family and relatives.

Data

National Education Panel Study (NEPS)

Our main analysis uses Starting Cohort 4 from the German NEPS (see Blossfeld et al. 2011). The NEPS provides a representative sample of schools in Germany using a stratified two-stage sampling procedure (for more details on the study design and sampling, see Skopek et al. 2013). We use data from the first two waves, both of which were conducted in grade 9 during the 2010–2011 academic year, when schooling was still compulsory for this cohort ($N = 14,540$ overall in regular schools). In addition to the standard version of the NEPS, which includes the year and month of birth, we use a restricted-access version that includes the week of birth for Starting Cohort 4. Because the week of birth was collected in later waves, it is available for about 85% of our main sample.

The NEPS interviews children and parents separately. The interviews contain rich information on parental education and occupations, migration background, income, and the household size and composition, which we use to assess the importance of potential confounders. Furthermore, the parents state the year and month when a child first entered formal childcare.¹⁰ We exclude children from the first-stage estimation for whom parents did not report a childcare starting age, which is the dependent variable in this regression.¹¹ We classify parents as possessing low/medium education if parents' highest level of education is less than upper secondary (*Fachabitur/Abitur*). Moreover, we classify parents as nonnatives if the mother's native tongue is not German.¹²

The NEPS provides standardized test scores for assessing children's competencies in different dimensions. We use these standardized skill measures for German language, STEM (science, technology, engineering, mathematics) subjects, and general cognition. The NEPS additionally collects information on the Strength and Difficulties Questionnaire (SDQ) and the Big Five personality traits. These and similar abilities go by different names in the literature (e.g., socioemotional skills and/or regulation). Following Heckman (2008), we refer to them as *noncognitive skills*. We again standardize

¹⁰ About 60% of parents completed the parental interview. Hence, we refer to the “full” and “parent” samples in the Results section, where we also provide evidence that our estimated treatment effects are not biased by selective nonresponse.

¹¹ In our sample, 8% of parents did not answer this question, and importantly, this share does not differ significantly across birth months (with a p value above .2). Given that 90% of children already attend childcare by age 5 and that children can still start childcare at this age, not reporting a childcare starting age most likely indicates that the children did not attend at all. Unfortunately, the survey does not include a separate item for whether a child ever attended childcare.

¹² Because the alternative form of care at that time was largely maternal care, children of nonnative parents would mainly be exposed to a foreign language in the counterfactual care situation. We are precisely interested in the treatment effect for this group, particularly in terms of language skills, and thus we prefer this linguistic definition of nativity.

each score to have a mean of 0 and a standard deviation of 1 (see online appendix B for further details).

Our analysis focuses on children who were born between July 1994 and June 1996. Because we observe these children in grade 9 in the 2010–2011 academic year, children born between July 1994 and June 1995 either repeated a grade or delayed their school entry; children born between July 1995 and June 1996 represent the regular school cohort.¹³

Administrative Data

We use two administrative data sets that cover the universe of children in two states: the Bavarian school census and school entrance examination data from Schleswig-Holstein.¹⁴ Both data sets are suitable for our analysis within a two-sample instrumental variables approach (see the Empirical Strategy section) given that they cover the same birth cohorts as the NEPS and that they yield more precise reduced-form estimates because of the larger sample. To alleviate concerns about the lack of comparability between these two states and the rest of West Germany, we additionally use data from the German Microcensus, which annually provides a 1% sample of the population. Table A3 in the online appendix compares basic sociodemographic characteristics correlated with children's skills. Although Bavaria and Schleswig-Holstein are somewhat more rural, they are almost indistinguishable from the rest of West Germany along most other sociodemographic characteristics.¹⁵

The first outcome variable in the administrative data is children's track choice. Germany's secondary school system consists of a basic vocational track (*Hauptschule*), an intermediate vocational track (*Realschule*), and an academic track (*Gymnasium*). Students are tracked into these types of secondary schools after completing primary school, typically at age 10. The tracking decision is based on teachers' assessments of children's academic performance, but parents have some discretion to enroll their children in a higher track than recommended in some states (for details, see Dustmann et al. 2017). Hence, children's track choice is an alternative measure of their skills (Schneeweis and Zweimüller 2014).

To examine track choice at age 15, the last year of compulsory secondary schooling, we use data from the Bavarian school census for the 2010–2011 academic school year ($N = 102,523$). The school census covers the full population of students in Bavaria. The

¹³ In Table A2 in the online appendix, we show that starting childcare earlier does not affect the probability of attending grade 9 at age 15. Because the proportion of children retained is fairly constant between adjacent cohorts of children, it is reasonable to assume that the children from the July 1994–June 1995 cohort who repeated a grade would be of similar academic quality to the children from the July 1995–June 1996 cohort. Moreover, we control for cohort differences in ability and interact the running variable by cohort, thereby accounting for ability differences between cohorts. Thus, including children who repeated a grade should not bias our estimates.

¹⁴ The NEPS data-use regulations prohibit reporting results for single states, and we thus can show our first-stage relationship only if we combine data for two states. Given this constraint and the limited availability of such data, we combine the data on school entrance examinations from Schleswig-Holstein (the only West German state for which such data are available) with school census data from Bavaria (the largest state for which school census data are available).

¹⁵ We later show that our first-stage estimates also hold for these two states. To alleviate concerns about the comparability of samples resulting from differences in urbanicity, we also show that the first-stage estimates do not differ by urbanicity.

data include information on month and year of birth, the attended track, and children's migration background, but not on any other sociodemographic characteristics or test scores.¹⁶

Additionally, we use data from school entrance examinations in Schleswig-Holstein, where these examinations are compulsory for all children shortly before entering primary school. These exams assess children's health, socioemotional development, language and motor skills, and ultimately school readiness. The standardized tests and assessments are all conducted by public health pediatricians. These data provide a comprehensive picture of child development.¹⁷ In some districts, parents additionally complete the SDQ. The school readiness recommendation is not a mechanical function of the medical diagnoses: pediatricians are allowed to weight the information differently depending on children's development and socioeconomic background. For instance, pediatricians may weight language skills differently by migration background. Ultimately, the recommendation is not binding.

Information on parental background characteristics is available for children born from July 1995 onward, and we therefore exclude children born prior to July 1995. For the subgroup analysis, we consider the following: (1) parental education, with parents classified as possessing low/medium education if their highest level of education is less than upper secondary;¹⁸ (2) the child's gender; and (3) migration background, with mothers classified as nonnatives if they were born abroad, given the absence of information on the mothers' native tongue or the primary language spoken at home.

Table A4 in the online appendix provides a concise overview of the different data sets.

Empirical Strategy

Enrollment in Childcare

Parents can theoretically enroll their children in at least three ways, as illustrated in panel a of Fig. 1.¹⁹ First, parents may enroll their children when they reach their legal entitlement age (i.e., the month they turn 3). This practice implies that children enroll continuously during the childcare year. If all parents enrolled their children exactly on the day their child turned 3, the average age at childcare entry would correspond to the horizontal line at age 3.

¹⁶ The scientific-use files contain only the quarter of birth. We negotiated access to month of birth, compromising on more-detailed migration characteristics. The scientific-use file contains the language spoken at home, children's nationality, and country of birth. Our version includes an indicator whether any of these three characteristics were non-German.

¹⁷ Pediatricians assess children's language based on children's vocabulary, articulation, and hearing problems. Motor skills are assessed through different exercises, including jumping across a line, standing on one leg, and jumping on one leg. To assess behavioral problems, pediatricians make a clinical assessment based on the child's behavior and parental information during the medical screening.

¹⁸ Because parents voluntarily provide the information, parental education is missing for about 40% of children.

¹⁹ Depending on the state, the school and childcare years start either in August or September. Without loss of generality, we focus on August as the starting point for ease of illustration.

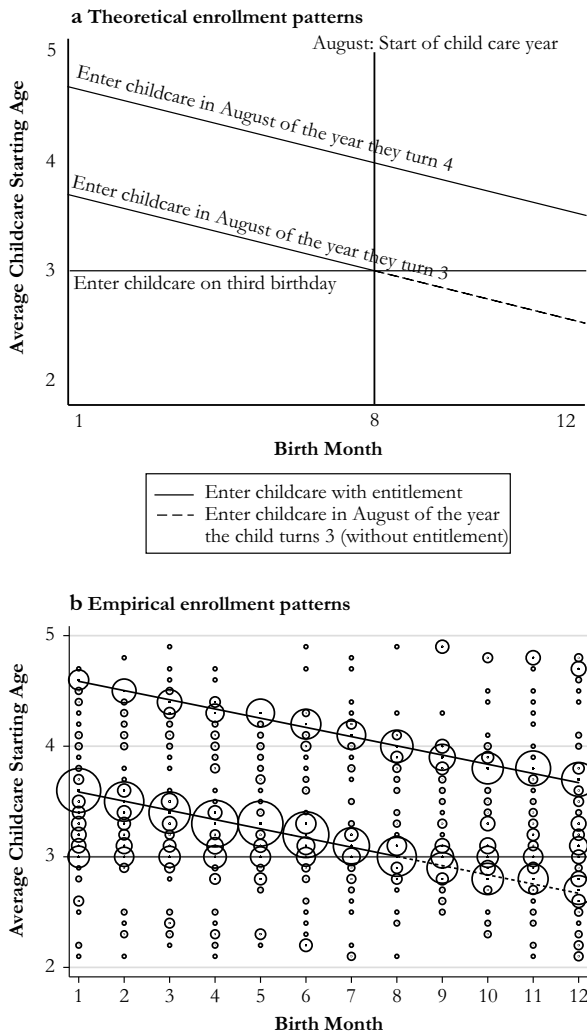


Fig. 1 Childcare enrollment patterns. Panel b uses the NEPS Starting Cohort 4 and proportionally weights the data points according to the number of children enrolling at a given age, by birth month. *Source:* Authors' own compilations.

Second, parents may enroll their children at the start of a childcare year after attaining legal entitlement. Sending the children after the summer holidays “has been customary in Germany for decades” (Frankfurter Allgemeine Zeitung (FAZ) 1996) and “independent of birth month” (FAZ 1997). This form of enrollment is fairly practical: when the oldest children leave childcare to start school in August, the vacant places become available for the next cohort of children to start childcare. This pattern would lead to a negative linear relationship between birth month and childcare starting age depicted by the downward-sloping lines. The shifts between the lines are explained by children enrolling the year they turn 4 or 3.

Third, children born between August and December attain their legal entitlement shortly after the beginning of the childcare year but still in the same calendar year. Childcare centers

were legally allowed to accept younger children if slots were available, and as discussed earlier, many districts supplied more slots than entitled children used. Parents may hence try to enroll their children at the beginning of the childcare year to allow their children to start childcare jointly with the rest of the group (see the dashed line).

Overall, which of these enrollment patterns dominates remains an open question.²⁰ We therefore examine the empirical evidence on enrollment using the NEPS survey. Overlaying the empirical evidence over the theoretical regimes, Panel b of Fig. 1 presents evidence supporting all three enrollment regimes. Three major patterns emerge. First, the concentration of observations along the diagonal lines shows that the majority of children enter childcare at the start of the school year. Second, few observations lie on the horizontal line at age 3. Thus, only few children start childcare the month they become legally entitled. Third, almost only children born between September and December systematically start childcare before their third birthday.²¹

Taken together, these results document that children typically start childcare in the calendar year in which they turn 3 at the start of the academic year. Hence, a substantial share of children born in December start in the summer before turning 3, whereas children born in January start in the summer after turning 3.

January-born children do not start childcare the summer before they turn 3 because of municipal statutes regarding cutoffs for early entrance to childcare. Because these statutes are not systematically available, we collected data on local rules determining early access to childcare. First, we contacted all 11 municipalities with at least 500,000 inhabitants to request information on their cutoff rules. Seven of eight responding municipalities use cutoffs that systematically prohibit January-born children from starting childcare in the summer before they turn 3; in the remaining city, childcare centers make entry decisions and may still use cutoff rules. Second, we also obtained information from several smaller municipalities, and a similar picture emerges. According to childcare officials, these cutoff rules allow childcare centers to fill all available slots, provide a communicable and clear-cut decision rule for early entrance, and ease forecasting the demand for slots. Although any cutoff would be viable to achieve these goals, the end of the calendar year seems to serve as a psychological focal point.

This enrollment practice generates a discontinuity in the average childcare starting age between December and January. Averaging over birth months and weeks across cohorts, panels a and b in Fig. 2 show that children born in December start childcare substantially earlier than children born in January of the subsequent year. The average age at starting childcare jumps discontinuously between December and January by just

²⁰ Unfortunately, official figures on childcare starting age are not available. However, the Federal Ministry for Family Affairs, Senior Citizens, Women and Youth reported that on December 31, 1999, 50% of children aged 3, 82% of children aged 4, and 90% of children aged 5 were enrolled in childcare (BMFSFJ 2005). Assuming that these children started in August/September, their average starting age was 3.7 years, which closely compares to 3.4 years in our data.

²¹ Figure A3 in the online appendix illustrates these specific enrollment patterns. Panels b and c together imply that about 40% to 50% of children born between October and December start childcare before or on their third birthday. This share compares very similarly to the childcare attendance rates that Bauernschuster and Schlotter (2015: figure 3) reported. Using data from the SOEP on actual attendance, they showed that about 45% to 60% of children born between October and December attend childcare in the first spring after turning 3; these children therefore must have started childcare either before or shortly after turning 3, consistent with our findings both qualitatively and quantitatively. Unfortunately, the SOEP data do not provide information on the month and year when children started childcare.

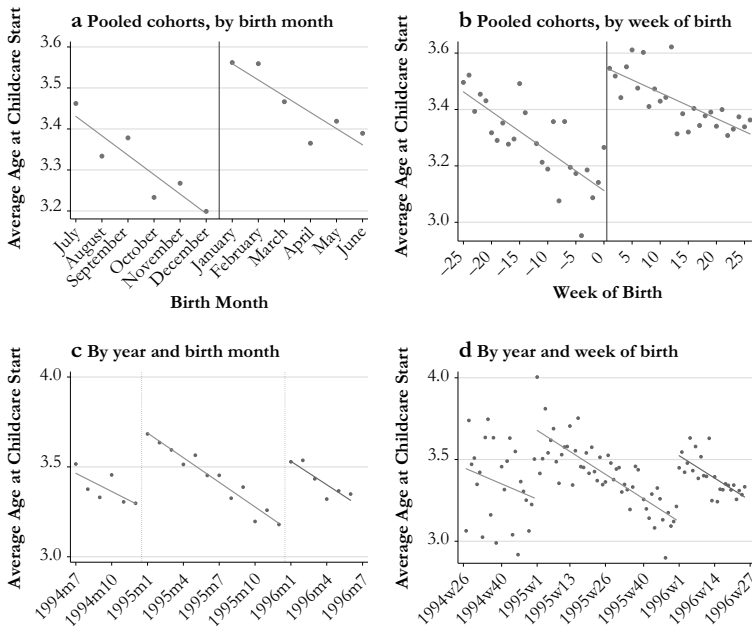


Fig. 2 Average childcare starting age. *Source:* NEPS Starting Cohort 4.

over 0.4 years (i.e., about 5 months). The discontinuous jump is of similar size for both December–January windows in our sample (see panels c and d).

To alleviate concerns about recall bias, we also examine childcare starting age in an alternative data set. The NEPS Starting Cohort 2 covers children born between 2005 and 2006, who were first surveyed in 2009, shortly after starting childcare. We observe a similar discontinuity in childcare starting age between December and January (see Fig. A4, online appendix). Again, this discontinuity is driven by the majority of parents enrolling their children at the start of the childcare year in August/September. We next describe how we use this relationship in our empirical methodology.

Fuzzy Regression Discontinuity Design

We model the effect of childcare starting age ($kgage$) on children's skill measures as follows:

$$y_i^k = \alpha_c^k + \beta^k kgage_i + f_c^k(M_i) + \phi^k \mathbf{X}_i + \varepsilon_i^k, \quad (1)$$

where the dependent variable y_i^k denotes the skill measure k for individual i . Because we observe two school entrance cohorts of children in grade 9, we include a cohort fixed effect (α_c^k). To maximize precision while avoiding the June/July school entrance cutoff,²² we restrict our sample to children born between August

²² In Germany, children start school at age 6, and the school entrance cutoff during the period of analysis was at the end of June (see Faust 2006). At that time, less than 3% of children started school earlier, and the share of children starting earlier is even lower in December and January (see Fig. A5, online appendix).

and May.²³ Given the importance of school starting age and relative age in class for later outcomes (Bedard and Dhuey 2006; Black et al. 2011), which may link children's birth months with their skills, we control for the running variable (date of birth) using a control function, $f_c^k(M_i)$, that also interacts month of birth, M_i , with the December/January threshold. In our preferred specification, we use a linear control function (which we also interact by cohort) to isolate the effect of starting childcare earlier from age effects. To alleviate concerns that a linear control function might not be flexible enough—for example, because of nonlinearities in brain development—we also test the sensitivity of our results with respect to changes in the specification of $f_c^k(M_i)$. Because the tests are administered at different times, we control for age at testing (measured in months) linearly.²⁴ $\mathbf{X}_i$ represents a vector of child and parent characteristics, and ε_i^k is an error term.

A simple ordinary least squares (OLS) regression of Eq. (1) will yield biased coefficients because parents are likely to selectively send their children to childcare earlier, or later, depending on (unobserved) parental preferences and children's skills. To solve the endogeneity of $kgage$, we use birth month to construct our instrument. For simplicity, we recode month of birth so that the December/January threshold lies in the center of the interval $M_i = (-4.5, \dots, 4.5)$. We define the indicator variable *before* as

$$before_i = I(M_i < 0). \quad (2)$$

Hence, *before_i* equals 1 for individuals born between August and December, and 0 otherwise. Our first-stage equation is

$$kgage_i = \alpha_{1c} + \delta_1 before_i + f_{1c}(M_i) + \phi_1 \mathbf{X}_i + \varepsilon_{1i}. \quad (3)$$

The parameter δ_1 identifies the discontinuous jump in average childcare starting age at the December/January threshold conditional on date-of-birth effects. Because the probability of starting childcare earlier does not switch from 0 to 1 between December and January (panel b, Fig. 2), our approach resembles a fuzzy regression discontinuity design. To compute the Wald estimator, we estimate the reduced form:

$$y_i^k = \alpha_{2c}^k + \delta_2^k before_i + f_{2c}^k(M_i) + \phi_2^k \mathbf{X}_i + \varepsilon_{2i}^k, \quad (4)$$

where δ_2^k denotes the average difference in skill k between children born before and after the December/January threshold. Using one data set, the Wald estimator corresponds to the two-stage least squares (2SLS) estimator. Given monotonicity and exogeneity, the Wald estimator for the effect of our continuous treatment variable

²³ In section C of the online appendix, we report all main results for a smaller, three-month window from October to March. The results remain the same, although they are estimated less precisely.

²⁴ For the effect of childcare starting age to be isolated from effects of age at testing, age at testing must vary within birth months and ideally overlap between birth months. These requirements are met in the NEPS tests and in the school entrance examinations because each test was administered to children at different dates; the NEPS tests were spread out over three months, and the school entrance examinations took place over five months. Figures A7 and A6 in the online appendix show the resulting variation in age at testing within and between birth months in both data sets. For a robustness check, we also used monthly dummy variables instead of linear age at testing; the results, available upon request, were the same.

identifies an average causal response, which computes a weighted average of several local average treatment effects (LATE; Angrist and Pischke 2009). Because we observe childcare starting age only in the NEPS, we cannot compute the 2SLS estimator using the administrative data sets. However, given that we observe month of birth in all data sets, we can compute the two-sample instrumental variables estimator (Angrist and Pischke 2009), dividing the reduced-form coefficient of any sample by the first-stage coefficient from the NEPS sample.

The approach relies on the assumption that the first stage applies equally to the different samples. As shown previously, Bavaria and Schleswig-Holstein are similar to the rest of West Germany regarding socioeconomic characteristics, apart from urbanicity (see Table A3, online appendix). The first-stage relationship holds similarly when data are pooled from Bavaria and Schleswig-Holstein only.²⁵ If anything, the first-stage relationship is slightly stronger in these two states than for West Germany as a whole, leading to an overestimate of the treatment effect using the overall first-stage coefficient. To alleviate concerns that differences in urbanicity bias our estimates, we reestimate the first stage separately for rural districts only.²⁶ The results, presented in Table A6 (online appendix), show no differences between rural and urban districts. Taken together, these pieces of evidence support our approach to combine the first-stage estimates from the NEPS with the reduced-form estimates for these two states.

Results

Instrument Validity and First-Stage Estimates

We perform two tests to examine whether the instrument is as good as randomly assigned. First, following McCrary (2008), we examine whether births are smoothly distributed around the December/January threshold. We observe no discontinuity in the frequency of births around December/January when using month and week of birth (see Fig. A9, online appendix), supporting that the running variable is not systematically manipulated.

Second, given the ongoing debate on selective seasonal birth patterns,²⁷ we perform balancing tests. We examine whether predetermined characteristics that predict children's skills differ discontinuously between children born before the December/January threshold and those born after it. Table 1 reports mean characteristics of children born

²⁵ Ideally, we would estimate the first stage separately for Bavaria and for Schleswig-Holstein. Unfortunately, as mentioned earlier, state-specific analyses are prohibited with the NEPS data, but we can combine at least two states for the analysis. Figure A8 (online appendix) presents the first stage using only observations from Bavaria and Schleswig-Holstein. Table A5 (online appendix) provides the corresponding regression coefficients.

²⁶ For these regressions, we merge additional information on the population density from the German Federal Statistical Office to the NEPS data. We classify a district as rural if the population density (i.e., the population per square kilometer) is below the median of the population-weighted population density, which was 377.5 in 1997. We picked 1997 because this was the year that the cohort children began to enter childcare.

²⁷ Most prominently, Buckles and Hungerman (2013) found seasonal differences in important observable characteristics, such as mothers' age and education, in the United States. In contrast, Fan et al. (2017) concluded in their cross-country study that seasonality is not omnipresent and showed that seasonality effects for the United States vanished when they appropriately controlled for race.

Table 1 Descriptive statistics: Covariate balancing

	Mean Values		<i>t</i> Test	Pei et al. (2019) Procedure		
	Before = 1	Before = 0		Difference	<i>p</i> Value	<i>p</i> Value
	(1)	(2)				(7)
			Difference (Δ)	Standardized Δ		
	(1)	(2)	(3)	(5)	(6)	(7)
A. Based on Children's Information						
Child male	.512	.498	.015	.029	.036	.081
Child age at testing (in 2010)	15.35	14.99	.358	.809	.002	.186
Mother nonnative speaker	.196	.189	.008	.020	.001	.986
Mother born abroad	.216	.205	.011	.028	.005	.765
Partner nonnative speaker	.151	.135	.016	.046	.008	.561
Partner born abroad	.223	.204	.019	.047	.011	.516
Mother's education						
None/lower secondary	.188	.169	.020	.051	-.007	.639
Medium secondary	.322	.319	.004	.008	.003	.872
Upper secondary	.135	.151	-.016	-.046	-.004	.786
Any tertiary	.101	.105	-.003	-.011	-.007	.555
Partner's education						
None/lower secondary	.208	.203	.004	.011	-.013	.419
Medium secondary	.229	.217	.012	.029	.031	.068
Upper secondary	.119	.138	-.019	-.058	-.037	.006
Any tertiary	.127	.129	-.001	-.004	.005	.737
Mother's occupational status						
1st quartile	.124	.106	.018	.056	.013	.316
2nd quartile	.162	.162	.000	.000	.014	.338
3rd quartile	.146	.155	-.008	-.023	.004	.807

Table 1 (continued)

	Mean Values		<i>t</i> Test	Pei et al. (2019) Procedure		
	Before = 1 (1)	Before = 0 (2)		Difference (Δ) (3)	Standardized Δ (5)	<i>p</i> Value (7)
4th quartile	.181	.196	-.016		-.040	.060
Father's occupational status						
1st quartile	.154	.145	.010		.027	.972
2nd quartile	.172	.161	.011		.031	.736
3rd quartile	.153	.169	-.016		-.044	.451
4th quartile	.162	.168	-.006		-.016	.616
<i>N</i>	4,766	4,942				
B. Based on Parents' Information						
Parent's age	44.80	44.73	.065		.012	.537
Mother nonnative speaker	.133	.123	.010		.030	.255
Mother born abroad	.144	.130	.014		.042	.012
Living arrangements						
Married/cohabiting	.765	.765	.000		.001	.987
Divorced/separated	.175	.182	-.007		-.017	.625
Widowed/single	.060	.053	.007		.030	.404
Household size	4.020	4.055	-.035		-.016	.942
Mother's education						
None	.017	.016	.002		.012	.569
Lower secondary	.105	.090	.015		.049	.317
Medium secondary	.477	.474	.003		.005	.160
Upper secondary	.142	.143	-.001		-.002	.357

Table 1 (continued)

	Mean Values		<i>t</i> Test				Pei et al. (2019) Procedure	
	Before = 1 (1)	Before = 0 (2)		Difference (Δ) (3)	<i>p</i> Value (4)	Standardized Δ (5)	Difference (6)	<i>p</i> Value (7)
Any tertiary	.162	.180	-.018		.073	-.047	.007	.720
Mother's work status								
Employed full-time	.323	.320	.003		.812	.006	-.006	.815
Employed part-time	.516	.529	-.013		.327	-.026	-.023	.383
Not employed	.162	.152	.010		.298	.028	.029	.134
Partner nonnative speaker	.140	.132	.009		.427	.025	.017	.433
Partner born abroad	.147	.147	.001		.950	.002	.006	.788
Partner's age	46.39	46.42	-.028		.875	-.005	-.391	.283
Partner's education								
None	.013	.008	.005		.119	.049	.001	.930
Lower secondary	.098	.083	.015		.092	.053	.000	.997
Medium secondary	.446	.446	.000		.993	.000	.005	.876
Upper secondary	.183	.178	.004		.724	.011	-.009	.709
Any tertiary	.252	.276	-.025		.075	-.056	-.004	.892
Partner's work status								
Employed Full-time	.772	.780	-.008		.533	-.020	-.023	.385
Employed Part-time	.120	.125	-.005		.647	-.014	.013	.523
Not employed	.107	.094	.013		.173	.043	.010	.612
Family income								
1st quartile	.110	.109	.002		.813	.005	.004	.777
2nd-3rd quartile	.244	.249	-.005		.562	-.012	.020	.255

Table 1 (continued)

	Mean Values		<i>t</i> Test		Pei et al. (2019) Procedure	
	Before = 1	Before = 0	Difference (Δ)	<i>p</i> Value	Standardized Δ	<i>p</i> Value
	(1)	(2)				
4th quartile	.120	.118	.002	.803	.005	.866
<i>N</i>	2,550	2,701				
C. Official Childcare Statistics (for children aged 3–6)						
Childcare spots per child in 1994	.862	.866	-.004	.399	-.022	.028
Childcare spots per child in 1998	1.035	1.037	-.002	.659	-.012	.024
Children per childcare worker	7.697	7.635	.061	.091	.045	.101
<i>N</i>	4,766	4,942				

Notes: Descriptive statistics using a five-month window around the December/January threshold. *Before* = 1 indicates that a child is born between August and December, and = 0 otherwise. The standardized difference of variable *x* in column 5 is defined as $\Delta_x = \frac{\bar{x}_1 - \bar{x}_0}{\sqrt{(\hat{\sigma}_1^2 + \hat{\sigma}_0^2)/2}}$, where $\hat{\sigma}_x^2$ represents the sample variance of *x* for children born before the January/December threshold (*before* = 1) and after the threshold (*before* = 0); see Rubin (2001). The results in columns 6 and 7 are based on the Pei et al. (2019) procedure and report the coefficient and *p* value for the *before* dummy variable from a regression of each covariate on the running variable and the *before* dummy variable analogous to our preferred specification (Eq. (4)) (without the *X* vector). For categorical variables, we omitted the missing category. The number presented in panel C is based on information on the number of children aged 3–6 in each district and the number of slots provided for children aged 3–6 in each district.

Source: Own calculations based on the NEPS Starting Cohort 4 (panels A and B) and official statistics from the German Federal Statistical Office (panel C).

Table 2 First-stage estimates of the effect of being born before December 31 on childcare starting age

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Five-Month Window ($N = 5,243$, month of birth)	-0.402*** (0.041)	-0.408*** (0.041)	-0.403*** (0.041)	-0.407*** (0.041)	-0.421*** (0.040)	-0.420*** (0.040)	-0.472*** (0.064)
First-stage F	98.07	101.18	98.19	100.87	111.65	111.07	55.08
B. Three-Month Window ($N = 3,190$, month of birth)	-0.382*** (0.054)	-0.390*** (0.054)	-0.384*** (0.054)	-0.393*** (0.054)	-0.428*** (0.054)	-0.431*** (0.054)	-0.436*** (0.102)
First-stage F	49.39	51.49	49.83	52.03	62.23	63.09	18.45
C. Five-Month Window ($N = 4,512$, week of birth)	-0.361*** (0.043)	-0.368*** (0.043)	-0.361*** (0.043)	-0.368*** (0.043)	-0.383*** (0.043)	-0.382*** (0.043)	-0.381*** (0.064)
First-stage F	69.82	72.29	69.69	72.13	80.30	79.71	35.85
Running Variable ^a	Linear	Linear	Linear	Linear	Linear	Linear	Quadratic
Interacted With Cutoff	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interacted With Cohort	No	No	Yes	Yes	No	Yes	Yes
Control Variables	No	Yes	No	Yes	Yes	Yes	Yes
District Fixed Effects	No	No	No	No	Yes	Yes	Yes

Notes: Control variables are as listed in Table 1, panel A. Robust standard errors are shown in parentheses.

Source: Own calculations based on the NEPS Starting Cohort 4, parent sample.

*** $p < .001$

within a five-month window before the December/January threshold and those born after, along with the group differences, corresponding t statistics, p values, and normalized differences. Additionally, following the procedure proposed by Pei et al. (2019), we test for discontinuous jumps in characteristics by regressing each characteristic on the instrument and the running variable.

Table 1 uses information on socioeconomic characteristics from the children's interviews (panel A) and the parents' interviews (panel B) as well as official childcare statistics (panel C). The results show no systematic differences or discontinuities between children born before and those born after the December/January threshold. For instance, the standardized differences are all below 0.1 and are even below 0.05 in 88.8% of cases. Normalized differences of magnitude 0.1 are common even in randomized experiments (Imbens and Rubin 2015:352). The only exception in Table 1 is child age, which is mechanically different across groups but still does not change discontinuously, as can be seen from the last two columns. That the instrument does not correlate with a rich set of observable characteristics also lends credibility to the assumption that our instrument is not confounded by unobservable characteristics, such as parents' ability to negotiate for a childcare slot.

We now turn to the size and robustness of the first-stage relationship. Table 2 shows the estimated first-stage coefficients, $\hat{\delta}_1$, from Eq. (3). To assess the robustness of the first-stage relationship, we present the results for different specifications and sample definitions. Panel A starts with a five-month-window around December/January using the month of birth. Column 1 shows the results of our baseline specification that controls only for a linear time trend in birth month interacted with the December/January threshold. In this specification, being born between August and December reduces the childcare starting age by 0.40 years, or 4.8 months. The coefficient is highly significant, with an F statistic of 98. In column 2, we include predetermined control variables from the children's survey, such as the child's gender and age, parental education, parents' place of birth, family status, mother's age, and household size. However, including these characteristics affects neither the estimated coefficient $\hat{\delta}_1$ nor its statistical significance. The same holds when we interact the control function with the cohort (see column 3) or combine both (see column 4). To control for regional differences in childcare coverage rates and quality, we also include 234 district fixed effects in columns 5 and 6. The point estimates increase to about 0.42 with and without interacting the running variable. Finally, column 7 additionally includes cohort-specific quadratic time trends interacted with the discontinuity. Allowing for this more flexible specification slightly increases the point estimate, and the coefficient is estimated less precisely. That neither the inclusion of a rich set of control variables, district fixed effects, nor a more flexible specification of the running variable substantially affect the point estimates compared with the basic specification in column 1 strengthens the assumption that month of birth is not systematically related to unobserved characteristics of the children or the families.

We next test whether narrowing the window around the discontinuity affects the first-stage estimates. Panel B in Table 2 presents the results including only children born between October and March. Across specifications 1 to 7, the point estimates are almost identical to those obtained with the wider window, although as expected, the estimates become less precise. We obtain a similar picture when using week of birth as the running variable, as shown in panel C. Given these consistent results, we proceed with our preferred specification (including district fixed effects and predetermined

characteristics, as reported in column 6), the larger window of five months, and month of birth in the main regressions to maximize the sample size.²⁸

Next, we examine potential heterogeneities in the first-stage relationship to ensure that the first-stage results also hold for the subgroups that we analyze separately. We categorize children into subgroups (by maternal education, migration background, and the child's gender), run a fully interacted model of the first stage, and report the estimated coefficients for the reference group and for the interaction term with the instrument. The instrument affects all groups very similarly: none of the interaction terms are significant (see Table A8, online appendix). This implies that our compliers are not a specific subgroup from the population of interest in terms of observable characteristics and that the first-stage relationship is strong and practically identical for the subgroups we analyze.²⁹

To further check the data quality and the robustness of our first-stage relationship, we use information on observable characteristics as reported by the children (panels A of Table A8, online appendix) and by the parents (panel B). Overall, the estimated first-stage relationships do not change substantially. The results show that information from the children survey and the parent survey lead to the same conclusions. We therefore use the children's information for our further analyses to maximize the sample size.

The Effect of Childcare Starting Age on Skills at Age 15

We begin our analysis of the effect of childcare starting age on skills and school track choice at age 15, given that these longer-run outcomes likely embody the most relevant path-determining outcomes of our analysis. We first focus on cognitive skill measures (language, STEM, and general cognition) and noncognitive skill measures (i.e., SDQ scores for peer problems and the Big Five personality traits). These skills ultimately determine children's track choice, which we analyze as an alternative, broader measure of ability. We start with a series of graphs reporting the reduced-form results. Figure 3 presents the overall picture by relating mean skill outcomes to month and week of birth. We observe rather smooth trends in average skill measures across birth months and weeks and, importantly, no discontinuity in outcomes around the December/January threshold.

Table 3 reports results from three different specifications. As shown in column 1, when we control only for the running variable, the reduced-form estimates of the effect are fairly small and statistically insignificant throughout. The estimates are close to 0 and range from a 0.06 SD decrease in neuroticism to a 0.04 SD increase in language skills. In columns 2 and 3, we include additional controls for the children's sociodemographic characteristics and district fixed effects, respectively. Including these control variables improves the explanatory power substantially, as reflected by the increasing R^2 . Reassuringly, the estimates are highly robust to these changes. Thus, we find no

²⁸ Using information on week of birth yields almost identical reduced-form results, see Table A7 in the online appendix.

²⁹ The first stage also does not differ by the presence of younger siblings, based on the number of children under age 14 in the household at the time of the first survey (results available upon request). As an alternative way to characterize the compliers, we follow Angrist and Pischke (2009) and recode our treatment variable into a binary indicator whether a child started childcare before the third birthday. The results reveal no clear pattern: most of the small differences across subgroups are not statistically significant (see Table A9, online appendix). We also calculate that 54.6% of children who attended childcare before their third birthday did so because they were born in the last quarter of the year. Hence, the compliers make up a substantial share of the children starting before age 3.

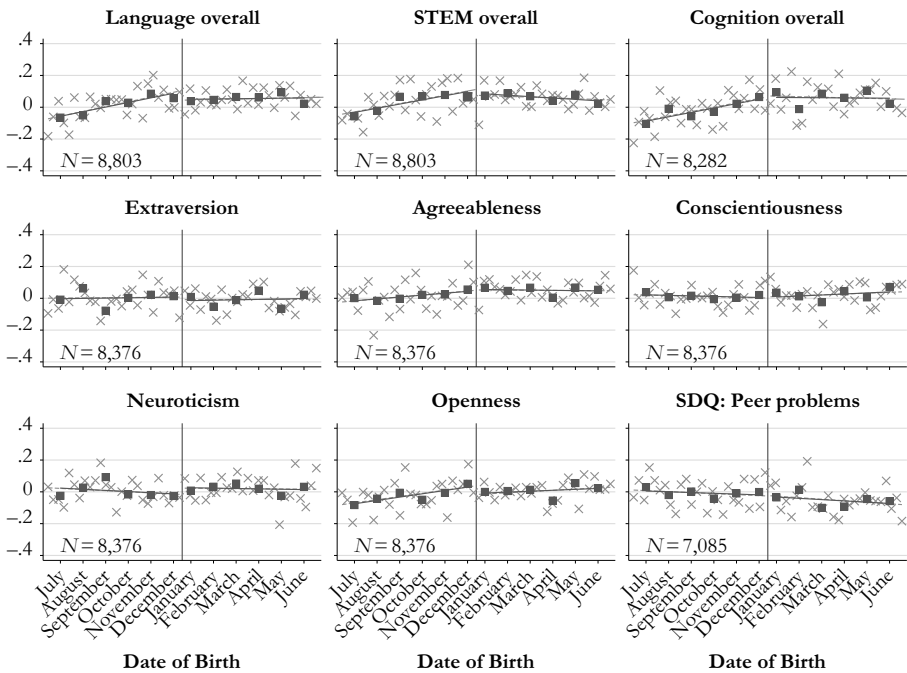


Fig. 3 Reduced-form effect on cognitive and noncognitive skills, based on the full sample. Dark squares and the dark solid line show linear trends based on the birth month. Gray x's and the dashed gray line show linear trends based on the week of birth. *Source:* NEPS Starting Cohort 4.

evidence on any of these outcomes, and our estimates are precise enough to rule out substantial effects on the order of 0.1 SD, consistent with findings by, for instance, Baker et al. (2008) and Drange and Havnes (2019).³⁰

In column 4, we report the Wald estimates, which divide the reduced-form estimates from column 3 by the corresponding first-stage estimate. The reported coefficients now refer to the effect of starting childcare one year later. The results show that the estimated effects remain small and statistically insignificant for all outcomes.³¹ In the following analyses, we do not scale up the results using the Wald estimator but report the reduced-form effects that stem from a five-month decrease in childcare starting age.³²

³⁰ According to a power analysis at the 10% significance level, we can detect a reduced-form effect of 0.1 SD with a probability of 90%. The calculation uses the STATA command *power oneslope*, specifying the number of observations ($N = 9,100$), the conditional standard deviation of the before dummy variable ($sd_x = 0.244$), and the conditional standard deviation of the language test score ($sd_y = 0.801$). We thank one referee for pointing out an inconsistency in our previous power analysis.

³¹ Figure A10 and Table A10 in the online appendix show the same patterns when we include only children from the parent sample. Because of the smaller sample size, the graphical patterns are noisier, and the estimates are less precise. However, the consistent results between both samples mitigate concerns about combining the first-stage results from the smaller parent sample with the reduced-form results for the full sample. Further, we reach the same conclusions when we replace the linear control function with a quadratic one (see Table A11, online appendix).

³² We focus on the reduced-form effects for two reasons. First, the Wald estimates are less precise, as indicated by the standard errors in Table 3. Second, the Wald estimates give the average marginal effect over the compliers' changes in childcare starting age. Because the underlying relationship between children's skills and their childcare starting age is potentially nonlinear, the effect should not be extrapolated to different changes in childcare starting age. Therefore, scaling up the effect does not help in interpreting the results.

Table 3 Reduced-form effect of early childcare attendance on test scores

	Reduced Form	Reduced Form	Reduced Form	2S-2SLS
Language Overall ($N = 9,132$)	0.037 (0.041)	0.063 [†] (0.037)	0.030 (0.035)	-0.073 (0.083)
R^2	.056	.218	.361	
STEM Overall ($N = 9,132$)	0.008 (0.041)	0.024 (0.038)	0.007 (0.035)	-0.018 (0.084)
R^2	.058	.231	.372	
Cognition Overall ($N = 8,529$)	-0.010 (0.043)	0.007 (0.042)	0.001 (0.041)	-0.001 (0.098)
R^2	.020	.050	.167	
Extraversion ($N = 8,659$)	0.004 (0.043)	0.012 (0.043)	0.004 (0.044)	-0.010 (0.102)
R^2	.000	.019	.045	
Agreeableness ($N = 8,659$)	0.011 (0.043)	0.022 (0.043)	0.036 (0.043)	-0.085 (0.101)
R^2	.001	.022	.051	
Conscientiousness ($N = 8,659$)	0.005 (0.044)	0.017 (0.043)	0.031 (0.043)	-0.074 (0.101)
R^2	.001	.048	.081	
Neuroticism ($N = 8,659$)	-0.062 (0.044)	-0.046 (0.043)	-0.039 (0.043)	0.092 (0.101)
R^2	.001	.054	.078	
Openness ($N = 8,659$)	0.037 (0.043)	0.062 (0.042)	0.057 (0.043)	-0.135 (0.100)
R^2	.001	.059	.086	
SDQ: Prosocial ($N = 7,211$)	0.003 (0.046)	0.024 (0.045)	0.029 (0.045)	-0.077 (0.117)
R^2	.001	.059	.097	
SDQ: Peer Problems ($N = 7,211$)	0.026 (0.048)	0.016 (0.048)	0.011 (0.048)	-0.028 (0.125)
R^2	.009	.023	.063	
Control Variables	No	Yes	Yes	Yes
District Fixed Effects	No	No	Yes	Yes

Notes: All dependent variables are standardized, with a mean of 0 and a standard deviation of 1. The specification of the running variable is as in column 6 of Table 2. Robust standard errors are shown in parentheses; standard errors on the two-sample two-stage-least squares (2S 2SLS) estimates are calculated using the delta method.

Source: Own calculations based on the NEPS Starting Cohort 4, full sample, for the five-month window around the December/January threshold.

[†] $p < .10$

Although we do not find an effect of starting childcare earlier, on average, the aggregate effect might mask substantial heterogeneity across particular groups. For

instance, we may expect that children whose mothers are not native German speakers would benefit from attending childcare earlier because of the higher exposure to the German language. We therefore next examine the effect of childcare starting age for specific subgroups. Because the analysis of the first stage (see Table A8, online appendix) did not reveal any significant differences across demographic groups, any potential effects for subgroups must result from differences in the reduced-form estimates.

Turning to the reduced-form results by maternal education, maternal native language, and child gender, Table 4 shows few statistically significant differences between children born before and after the December/January threshold. Although we find significant positive effects on language skills (0.20 SD) and cognition (0.21 SD) for children of nonnative mothers, a closer inspection of the corresponding graph shows that January-born children, who score particularly badly, drive this result. Applying a doughnut-hole specification (i.e., excluding December and January from the estimation), we observe no statistically significant difference in average language scores between nonnative children born before the December/January threshold and those born after.³³ We therefore conclude that the subgroup analysis does not reveal any meaningful differences for any of the considered outcomes.

Next, we examine the effect on children's track choice. If attending childcare earlier influences children's skills, this improvement might be reflected at different track choice margins depending on children's abilities: for children with medium to high ability levels, skill improvements should increase the probability of attending the academic track, *Gymnasium*. For children with low to medium ability levels, skill improvements might not suffice to attend the academic track. However, the improvements could lift such children from the basic track, *Hauptschule*, to the intermediate vocational track, *Realschule*. We therefore examine both margins.

We start with the probability of attending *Gymnasium*. First, using both the NEPS and Bavarian administrative data, we relate the proportion of students attending *Gymnasium* to year and month of birth (see Fig. A12, online appendix). We do not observe a discontinuity in the probability of attending *Gymnasium* at either of the December/January thresholds.³⁴ Reassuring for our two-sample instrumental variables approach, the figure shows that the share of students attending the academic track by year and month of birth is similar in both data sets. Hence, we proceed with the larger Bavarian sample for the subgroup analysis.

Table 5 provides the reduced-form coefficients for the effect on track choice controlling for a quadratic trend in month of birth (see Fig. A13 in the online appendix for the corresponding graphs). In panel A, we examine the probability of

³³ For the graphical analysis, see Fig. A11 (online appendix). The January dip is mostly driven by children of Turkish mothers. When we remove Turkish children ($N = 399$) from the sample, the treatment effects become small and statistically insignificant.

³⁴ Because we observe only children who attend grade 9, children born between July 1994 and June 1995 have either started school late or repeated a grade. Hence, they tend to have lower academic skills and a lower probability of attending *Gymnasium* compared with children in the same grade but born between July 1995 and June 1996.

Table 4 Reduced-form effect of early childcare attendance on test scores

	All	Mother's Education		Mother's Ethnicity		Child's Gender	
		Low	High	Native	Nonnative	Boys	Girls
Language Overall	0.028 (0.035)	0.037 (0.047)	0.025 (0.073)	−0.006 (0.037)	0.205* (0.099)	−0.013 (0.049)	0.054 (0.051)
<i>N</i>	9,131	4,652	2,360	7,424	1,707	4,597	4,534
STEM Overall	0.006 (0.035)	0.019 (0.047)	−0.032 (0.080)	−0.014 (0.039)	0.080 (0.084)	−0.028 (0.052)	0.048 (0.048)
<i>N</i>	9,131	4,652	2,359	7,424	1,707	4,597	4,534
Cognition Overall	−0.000 (0.041)	−0.017 (0.058)	0.017 (0.078)	−0.045 (0.044)	0.214† (0.113)	−0.051 (0.060)	0.038 (0.057)
<i>N</i>	8,526	4,320	2,218	6,947	1,579	4,296	4,230
Extraversion	0.010 (0.044)	−0.028 (0.063)	0.026 (0.091)	−0.001 (0.049)	0.042 (0.105)	−0.036 (0.061)	0.057 (0.065)
<i>N</i>	8,660	4,448	2,258	7,081	1,579	4,316	4,344
Agreeableness	0.038 (0.043)	0.017 (0.060)	0.004 (0.092)	0.024 (0.047)	0.084 (0.118)	−0.005 (0.063)	0.089 (0.060)
<i>N</i>	8,660	4,448	2,258	7,081	1,579	4,316	4,344
Conscientiousness	0.035 (0.043)	0.044 (0.061)	0.066 (0.089)	0.041 (0.048)	0.102 (0.115)	−0.016 (0.062)	0.088 (0.062)
<i>N</i>	8,660	4,448	2,258	7,081	1,579	4,316	4,344
Neuroticism	−0.038 (0.043)	−0.103† (0.062)	−0.005 (0.091)	−0.042 (0.048)	−0.002 (0.113)	−0.094 (0.062)	−0.005 (0.062)
<i>N</i>	8,660	4,448	2,258	7,081	1,579	4,316	4,344
Openness	0.051 (0.043)	0.000 (0.061)	0.108 (0.089)	0.041 (0.047)	0.088 (0.112)	0.055 (0.063)	0.068 (0.060)
<i>N</i>	8,660	4,448	2,258	7,081	1,579	4,316	4,344
SDQ: Prosocial	0.029 (0.046)	−0.005 (0.063)	−0.024 (0.102)	0.033 (0.051)	−0.056 (0.121)	−0.033 (0.070)	0.098 (0.061)
<i>N</i>	7,211	3,625	1,784	5,830	1,381	3,540	3,671
SDQ: Peer Problems	0.011 (0.049)	0.101 (0.069)	−0.153 (0.100)	−0.011 (0.055)	0.133 (0.118)	0.075 (0.075)	−0.025 (0.065)
<i>N</i>	7,211	3,625	1,784	5,830	1,381	3,540	3,671

Notes: All dependent variables are standardized, with a mean of 0 and a standard deviation of 1. The specification of the running variable is as in column 6 of Table 2. Robust standard errors are shown in parentheses.

Source: Own calculations based on the NEPS Starting Cohort 4, full sample, for the five-month window around the December/January threshold.

† $p < .10$

attending *Gymnasium*. We find no effect. Looking at nonnatives, the probability of attending *Gymnasium* is merely 0.6 percentage points larger for children born before the December/January threshold, and the difference falls short of statistical

Table 5 Reduced-form effect of early childcare attendance on track choice in Grade 9

	All	Native	Nonnative
<i>A. Gymnasium</i>			
All	-0.011 (0.009)	-0.014 (0.010)	0.006 (0.021)
<i>N</i>	102,523	90,892	11,631
Boys	-0.009 (0.013)	-0.014 (0.014)	0.024 (0.029)
<i>N</i>	52,922	46,895	6,027
Girls	-0.014 (0.013)	-0.014 (0.014)	-0.015 (0.032)
<i>N</i>	49,601	43,997	5,604
<i>B. Hauptschule</i>			
All	0.005 (0.010)	0.005 (0.010)	0.009 (0.029)
<i>N</i>	102,523	90,892	11,631
Boys	0.013 (0.014)	0.016 (0.014)	0.010 (0.040)
<i>N</i>	52,922	46,895	6,027
Girls	-0.004 (0.013)	-0.007 (0.014)	0.011 (0.042)
<i>N</i>	49,601	43,997	5,604

Notes: The dependent variables are binary variables. The specification of the running variable is as shown in column 7 of Table 2. Robust standard errors are shown in parentheses.

Source: Own calculations based on administrative data from the Bavarian school census for the five-month window around the December/January threshold.

significance. For this large data set, the estimates are again precise enough to rule out any substantial effect.³⁵

Turning to the effect on attending the basic track, *Hauptschule*, panel B reports the respective reduced-form estimates. As before, we find no effect for students overall. For nonnative students, the probability of attending the basic track is 0.9 percentage points larger for those born before the December/January threshold. This difference not only is statistically insignificant but also runs counter to the expected improvements in cognitive skills.

Does the Effect Fade Out Over Time?

We have provided new, comprehensive evidence that attending childcare earlier does not affect children's cognitive or noncognitive skills in grade 9 at age 15. We now examine whether early childcare ever had an effect by using data from school entrance examinations. Given that these examinations take place before children

³⁵ Because the data do not include additional information, we cannot control for additional child and parent characteristics. However, as Tables 2 and 3 show, our results from the NEPS do not depend on including additional control variables.

Table 6 Reduced-form effect of early childcare attendance on school entrance exams

	Mother's Education			Mother's Ethnicity		Child's Gender	
	All	Low	High	Native	Nonnative	Boys	Girls
A. Birth Cohorts July 1995 to June 1996							
Not school ready (mean = .052)	−0.007 (0.008)	−0.005 (0.012)	0.034 (0.030)	−0.007 (0.009)	0.033 (0.032)	0.006 (0.013)	−0.019† (0.010)
Speech difficulties (mean = .156)	−0.027† (0.016)	−0.007 (0.024)	−0.079 (0.049)	−0.032† (0.019)	−0.023 (0.043)	−0.011 (0.024)	−0.041* (0.020)
Behavioral difficulties (mean = .068)	0.008 (0.010)	0.007 (0.015)	0.024 (0.032)	0.003 (0.013)	0.026 (0.031)	0.023 (0.017)	−0.006 (0.013)
Motor skill difficulties (mean = .131)	0.007 (0.014)	0.005 (0.021)	−0.047 (0.042)	0.005 (0.017)	0.014 (0.040)	0.009 (0.023)	0.007 (0.015)
N	23,590	10,589	2,525	16,441	2,294	12,022	11,568
B. Birth Cohorts July 1995 to June 1998							
Not school ready (mean = .044)	−0.008* (0.004)	−0.012* (0.006)	−0.010 (0.013)	−0.008† (0.004)	−0.019 (0.016)	−0.002 (0.006)	−0.014* (0.005)
Speech difficulties (mean = .165)	0.003 (0.009)	0.002 (0.013)	−0.005 (0.026)	−0.004 (0.011)	0.007 (0.025)	0.013 (0.014)	−0.006 (0.012)
Behavioral difficulties (mean = .070)	0.007 (0.006)	0.005 (0.009)	−0.001 (0.019)	0.009 (0.007)	−0.025 (0.017)	0.010 (0.010)	0.005 (0.007)
Motor skill difficulties (mean = .144)	0.006 (0.008)	−0.021† (0.012)	0.021 (0.025)	0.003 (0.010)	−0.016 (0.023)	0.010 (0.013)	0.003 (0.009)
N	72,753	35,062	9,326	52,957	7,710	37,218	35,535

Notes: The dependent variables are binary variables. The specification of the running variable is as shown in column 7 of Table 2 and includes district fixed effects. Robust standard errors are shown in parentheses.

Source: Own calculations based on administrative school entrance examinations from Schleswig-Holstein for the five-month window around the December/January threshold.

†p < .10; *p < .05

enter school, examining preschool outcomes alleviates the concerns that any effects of absolute or relative age at school entry might contaminate our estimates at approximately age 15. Although we would expect positive initial effects to lead to further gains during the process of human capital formation if the skill accumulation process exhibits dynamic complementarities (Cunha and Heckman 2007), initial gains might also fade out over time.³⁶ To assess whether initial effects fade out by age 15, we next examine the effect earlier in life—just before entering school.

For our reduced-form analysis of the effect of childcare starting age on early skill measures, we use administrative data from school entrance examinations in Schleswig-Holstein. Panel A of Table 6 reports the results for children born between July 1995 and June 1996, the earliest cohort for which we observe parental characteristics. In panel B, we pool children born between July 1995 and June 1998 to gain precision in the subgroup analyses.

Controlling for a quadratic trend in month of birth as before, we report the effects on four outcomes: school readiness, language development, behavioral difficulties, and motor skill difficulties (see Fig. A14 in the online appendix for corresponding graphs). Column 1 of panel A shows no average effects on most outcomes, apart from a marginally significant reduction in speech difficulties. However, once we add the later-born children (panel B), the effect vanishes. Column 1 of panel B shows a significant 0.8 percentage point increase in school readiness for children born in December. The remaining columns reveal similar point estimates across all subgroups except boys. However, because these estimates are not robust to applying a doughnut-hole specification or modelling the running variable differently, we caution against interpreting the estimates as robust causal evidence. Overall, the visual evidence and corresponding regressions for both samples do not indicate any substantial effects on the analyzed outcomes.

Differences by Childcare Quality

We have shown that starting childcare earlier has no effect on several relevant cognitive and noncognitive skill measures at different points in children's lives. We don't know, however, what explains this finding and, in particular, why we find little to no evidence for positive effects on children from nonnative and low socioeconomic backgrounds. Understanding the mechanisms underlying the absence of an effect is crucial to design effective childcare policies.

Regarding potential learning mechanisms, suppose that children's skill development can be described as

$$y_t + (1 - t_C)q_H + t_C q_C = y_{t+1}, \quad (5)$$

where y_t and y_{t+1} denote the skill levels at the beginning and end of a period, during which children spend the portion t_C of their time in formal childcare and the

³⁶ Examples of partial fading-out include the Perry Pre-School trial, in which the initial IQ gains vanished by age 10. However, no fade out occurred for noncognitive skills: children still performed better in achievement tests later because of higher noncognitive skills (Heckman et al. 2013).

remaining portion at home. Further, q_j can be thought of as the quality of educational inputs in home care ($j = H$) and formal childcare ($j = C$). Starting childcare earlier corresponds to an increase of t_c by some time share p , and this intervention changes skill development by

$$\Delta = p(q_C - q_H). \quad (6)$$

Hence, the change in the exposure to both environments and the difference in the quality of educational inputs determine the treatment effect. Thus, either too little exposure or insufficient differences in the quality of educational inputs may explain our finding of no effect on children's skill development.

Children born in December spend about 400 more hours in formal childcare than children born in January,³⁷ and our literature review shows that this increase can make a difference for children's skill development.³⁸ Thus, insufficient differences in educational quality remain a potential explanation for our null findings. To probe this explanation, we increase the variation in this difference by examining additional subgroups. To capture educational quality at home, we use maternal education and migration background. To proxy childcare quality, we use the child-to-staff ratio at the district level, absent more-disaggregated information or information on other quality aspects.³⁹ We expect the largest gains for children from relatively low-quality home environments who enter high-quality childcare programs.

We focus on the school entrance examinations data from Schleswig-Holstein, which are collected when children still attend childcare and thus allow us to reliably identify the district of attendance.⁴⁰ In this sample, the child-to-staff ratio ranges from 5.8 to 18.5, with a standard deviation of 2.45. We define a high-

³⁷ For this calculation, we assume that children born in December attend childcare five months longer than children born in January and that they attend childcare for five days per week and four hours per day. The additional time spent in childcare then is 430 hours. Even if children missed out on one month (e.g., because of illness and/or holidays), we would arrive at 344 hours, which can be considered a lower bound.

³⁸ For instance, Ichino et al. (forthcoming) showed that for a mixture of full- and part-time attendance, one additional month significantly reduces cognitive skills. Four hundred hours, as in our setting, correspond to 2.5 months of full-time attendance. Furthermore, high-quality preschool programs of similar intensity have been found to substantially affect children's skill development (e.g., the Tulsa program; see Gormley and Gayer 2005), as have targeted programs (e.g., Head Start; see Garces et al. 2002). Additionally, Felfe and Zierow (2018) found that switching from half-day to full-day universal programs in Germany increased socioemotional problems by 0.18 SD. Because only 10% of the children in their sample responded to the increased access to full-day slots and full-day slots provide roughly 16 additional hours of care per week, their reduced-form estimates compare children whose average childcare attendance differed by at most 220 hours.

³⁹ The quality of formal childcare ultimately depends on the quality of interactions between children and staff. The child-to-staff ratio determines the types of interactions that are feasible and often serves as one indicator of "structural" childcare quality (e.g., Blau and Hagy 1998; Hofferth and Wissoker 1992). Furthermore, Rege et al. (2018) showed that the child-to-staff ratio explains roughly 30% of the variance in school readiness between childcare centers. Because we are not estimating the causal effect of quality or the child-to-staff ratio but we inspect effect heterogeneities along this dimension, such a proxy should suffice as an indicator of care quality. To compute the ratio, we use administrative data from the German Federal Statistical Office at the district level on (1) the number of formal childcare spots for children aged 3–6 and (2) the number of employed personnel (measured on December 31, 1998).

⁴⁰ The data sets covering the outcomes at age 15 measure the location only at the time of data collection but do not include information on earlier locations. Even short-distance moves across districts between attending childcare and age 15 would hence introduce measurement error in assigning the district of attendance.

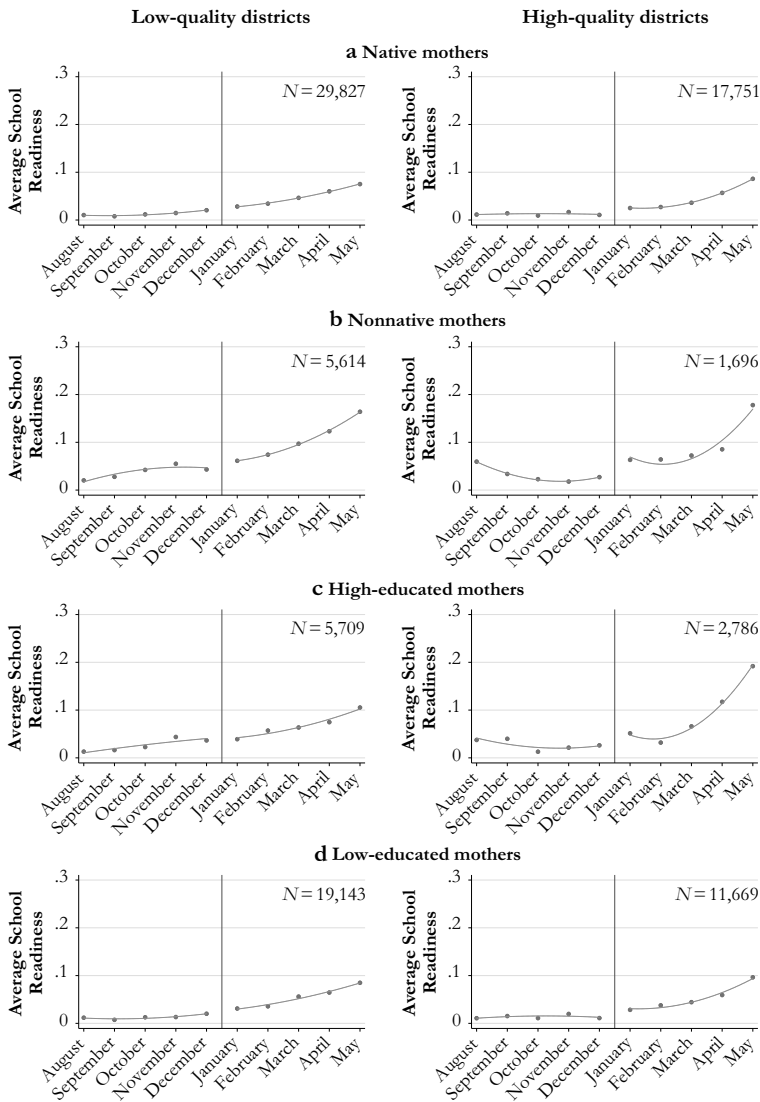


Fig. 4 Reduced-form effect on school readiness at age 6 for birth cohorts 1995 (July) to 1998 (June). *Source:* Administrative data from school entrance examinations in Schleswig-Holstein.

quality ratio as being below the population-weighted median of 11.2.⁴¹ Our discussion focuses on school readiness as the most comprehensive measure of overall skill development at this age. Because the proxies of quality appear rather crude, our sample size becomes relatively small for these combinations and

⁴¹ According to the regulatory standards, the typical ratio should be close to 11 (*Verordnung für Kindertageseinrichtungen Schleswig-Holstein*). Local decision-makers can implement lower ratios—say, for political reasons to attract parents who value higher quality. Moreover, the regulations do not state specific sanctions if the ratio is too high, and closures of childcare facilities seem politically infeasible because they would likely deny children access to childcare and/or negatively affect other facilities. This absence of a clear sanctioning mechanism may explain higher child-to-staff ratios in other districts.

because parents do not randomly select into districts (which may also differ along other dimensions), our results allow for only a tentative interpretation indicating directions for future research.

Figure 4 shows the reduced-form results for school readiness. The first column shows that children in lower-quality districts are not affected by starting childcare earlier, which the regression results confirm (see Table A12, online appendix). In contrast, children of nonnative mothers benefit from starting childcare earlier in high-quality districts (see panel b, Fig. 4). A similar picture emerges for children with low-educated mothers (panel d of Fig. 4), although the December/January difference is less pronounced. The regression results presented in Table A12 (online appendix) confirm that the strongest improvements in school readiness are for children of nonnative mothers living in districts with high-quality childcare, although the estimate is not statistically significant because of the limited sample size. We also observe a substantial and significant improvement in school readiness for children with low-educated mothers in high-quality childcare districts. The point estimates for these two groups are also robust when we exclude December and January from the estimation (see Table A13, online appendix). Our baseline specification indicates beneficial effects for children with native and high-educated mothers in high-quality childcare districts (see Fig. 4, panels a and c, respectively), although these effects are not robust to minor changes in the specification (see Table A13, online appendix). Regression results on the other available skill measures do not show systematic differences.

Discussion

Although previous studies reached no consensus on the average effect of starting childcare earlier, disadvantaged children tend to benefit. Our findings stand out from this pattern: we do not find any clear evidence that low-SES children benefit, on average, from starting childcare earlier. However, consistent with the broader literature, including that on targeted programs, we find some suggestive evidence that low-SES children benefit from attending childcare in high-quality districts in the short run. Whereas some studies have found negative effects for children from high-SES families, we do not find negative effects for this group. Because the studies differ along several important dimensions (e.g., the quality of care modes, the institutional setting, and the group of compliers), we are unable to single out a particular source driving the (partially) divergent findings.

To examine the role of parental resistance to early childcare, we compare our results with those of Cornelissen et al. (2018) showing that the gains from starting childcare early are concentrated among children of parents with high resistance. In contrast, because the compliers in our setting enter childcare before becoming legally entitled, they plausibly have a low resistance against starting childcare earlier and thus low gains from doing so.⁴² Hence, our null findings are consistent with those of Cornelissen et al.

⁴² To support this econometrically, we follow Heckman et al. (2006) and calculate the different weights of observations in an instrumental variable estimation. Because the approach requires a binary treatment, we recode our treatment variable to an indicator whether a child started childcare before the third birthday. Hence, we obtain the weights that our instrument assigns to children with different propensities to start childcare before the third birthday. Figure A15 in the online appendix shows that our instrument puts the most weight on individuals with a low resistance toward childcare and hardly any weight on children with a medium to high resistance to childcare.

(2018). Whereas their point estimates indicate that starting childcare early reduces the school readiness of low-resistance children, we show that these children do not exhibit any negative effects in the long run.

One limitation of our study is that we cannot distinguish between full- and part-time attendance, which is particularly relevant given that children aged 0–2 who are enrolled in formal childcare attend for about 30 hours per week (unweighted average across OECD countries as of 2014; see OECD 2017). Thus, future research should investigate in more detail whether the effects differ between full- and part-time attendance. Data limitations also prevent us from analyzing the long-run effects by different types of childcare quality, which promises to be an interesting avenue for future research. Further, our finding that starting childcare earlier has no effect on children's skills is limited to starting childcare at approximately age 3. Given the international trend of increasing childcare participation for children below age 3, an open question is how starting childcare between ages 0 and 2 affects children's skill development in the short and longer run.

Concerning the universality of our findings and the implications for current policy debates, our results refer to a setting where attending childcare around age 3 was less common than it is today and where almost all children attended part-time. Despite such differences, the underlying conditions of our study are relevant beyond the German context for policymakers currently considering whether to expand eligibility for younger children. First, the increase in time spent in childcare is sizable and similar to that of other programs. Children born in December spend about 400 additional hours in formal childcare than children born in January. This corresponds to an intensity of about 80 hours per month, which is almost identical to the new standard for the Head Start program set at 85 hours per month by 2021. Second, starting childcare five months earlier, around a child's third birthday, is a highly policy-relevant margin because the childcare coverage rates drop sharply around this age across most OECD countries compared with children aged 3 and older (see Fig. A1, online appendix). Third, the alternative care mode at the time was almost exclusively home care, which is still highly relevant for children in this age group.

Conclusion

In recent years, many countries have introduced policies that increase participation of younger children in childcare programs by lowering the childcare starting age. It is therefore important to understand whether starting universal childcare earlier affects children's skills. In this study, we use detailed survey and administrative data from Germany to investigate short- and longer-run effects of starting childcare earlier on a comprehensive set of skill measures. For identification, we exploit a fuzzy discontinuity in average childcare starting age between children born in December and January of the next year. Because this discontinuity does not affect children's age at school entrance, starting childcare earlier increases the duration of childcare attendance.

Our key finding is that children who start universal childcare earlier do not perform differently in terms of standardized cognitive test scores, measures of noncognitive skills, school track choice, or school entrance examinations. In addition, we find no evidence of skill improvements for children with less-educated or nonnative mothers. Our results are consistent across three data sets. The estimated treatment effect stems

from children who start childcare before they become legally entitled. These children plausibly have a low resistance against early childcare, and one would hence expect low gains or even negative effects for them. By analyzing the effect of starting childcare earlier for this group, we provide further evidence on the relationship between parental resistance to and children's potential gains from childcare.

To further explore our results, we explore the interplay between educational quality in the home environment and in childcare. This part of our analysis allows for only a tentative interpretation for three reasons: (1) parents living in districts offering different care quality are likely to differ systematically, (2) our measures of childcare quality are rather crude, and (3) our sample sizes become relatively small in subgroup analyses. Our results suggest that children with low SES benefit in terms of school readiness from starting high-quality childcare earlier. Differences in typical childcare quality by socioeconomic background have largely been neglected in the literature. Future research should focus more on this often unobserved relation between the quality of both the home and the formal childcare environments.

Our main policy implication is that simply easing access to childcare may not suffice to improve children's skill development, although we find no evidence that starting childcare earlier at around age 3 harms children's skill development. Instead, our results point toward the importance of improving the quality of childcare programs. Childcare programs of sufficiently high quality promise to foster children's skill development and improve mothers' labor market access, which taken together may alleviate the demographic challenges imposed by shrinking working-age populations in aging societies.

Acknowledgments We are grateful for helpful comments and suggestions received from four referees; Anna Adamecz-Völgyi, Stefan Bauernschuster, Jan Bietenbeck, Christian Dustmann, David Figlio, Jonathan Guryon, James J. Heckman, Mathias Huebener, Chris Karbownik, Patrick Puhani, Regina T. Riphahn, Claus Schnabel, Stefanie Schurer, Steven Stillman, Konstantinos Tatsiramos, and Rudolph Winter-Ebmer; from participants at the 2015 summer school of the DFG Priority Program 1764, the 2016 Ce2 workshop in Warsaw, the 2016 NEPS User Conference, the 1st IZA workshop on Gender and Family Economics, the 3rd network workshop of the DFG Priority Program 1764, the 2017 Essen Health and Labour Conference, the 2017 annual meeting of the Society of Labor Economics, and the 2017 annual meeting of the European Society of Population Economics; and seminar participants at the Northwestern Applied Micro Reading Group, Bayreuth University, Ludwig-Maximilians-Universität Munich, Lüneburg University, and RWI (Essen). This paper uses data from the National Educational Panel Study (NEPS): Starting Cohort 4–9th Grade, doi: <https://doi.org/10.5157/NEPS:SC4:4.0.0>. From 2008 to 2013, NEPS data were collected as part of the Framework Programme for the Promotion of Empirical Educational Research funded by the German Federal Ministry of Education and Research (BMBF). As of 2014, the NEPS survey is carried out by the Leibniz Institute for Educational Trajectories (LifBi) at the University of Bamberg in cooperation with a nationwide network. We thank the Ministry of Social Affairs, Health, Science and Equality of Schleswig Holstein, and Ute Thyen and Sabine Brehm specifically, for granting access to and providing valuable information on the school entrance examinations. Daniel Kuehnle acknowledges financial support by the DFG (grant number RI 856/7-1).

References

- Almlund, M., Duckworth, A. L., Heckman, J. J., & Kautz, T. (2011). Personality psychology and economics. In E. A. Hanushek, S. Machin, & L. Woessmann (Eds.), *Handbook of the economics of education* (Vol. 4, pp. 1–181). Amsterdam, the Netherlands: Elsevier.
- Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton, NJ: Princeton University Press.

- Baker, M., Gruber, J., & Milligan, K. (2008). Universal child care, maternal labor supply, and family well-being. *Journal of Political Economy*, 116, 709–745.
- Baker, M., Gruber, J., & Milligan, K. (2019). The long-run impacts of a universal child care program. *American Economic Journal: Economic Policy*, 11(3), 1–26.
- Bauernschuster, S., Hener, T., & Rainer, H. (2016). Children of a (policy) revolution: The introduction of universal child care and its effect on fertility. *Journal of the European Economic Association*, 14, 975–1005.
- Bauernschuster, S., & Schlotter, M. (2015). Public child care and mothers' labor supply—Evidence from two quasi-experiments. *Journal of Public Economics*, 123, 1–16.
- Bedard, K., & Dhuey, E. (2006). The persistence of early childhood maturity: International evidence of long-run age effects. *Quarterly Journal of Economics*, 121, 1437–1472.
- Bernal, R., & Keane, M. P. (2011). Child care choices and children's cognitive achievement: The case of single mothers. *Journal of Labor Economics*, 29, 459–512.
- Black, S. E., Devereux, P. J., & Salvanes, K. G. (2011). Too young to leave the nest? The effects of school starting age. *Review of Economics and Statistics*, 93, 455–467.
- Blau, D., & Currie, J. (2006). Pre-school, day care, and after-school care: Who's minding the kids? In E. A. Hanushek & F. Welch (Eds.), *Handbook of the economics of education* (Vol. 2, pp. 1163–1278). Amsterdam, the Netherlands: Elsevier.
- Blau, D. M., & Hagy, A. P. (1998). The demand for quality in child care. *Journal of Political Economy*, 106, 104–146.
- Blossfeld, H.-P., Roßbach, H.-G., & von Maurice, J. (2011). *Education as a lifelong process—The German National Educational Panel Study (NEPS)*. Wiesbaden, Germany: VS Verlag für Sozialwissenschaften.
- BMFSFJ. (2005). *Zwölfter Kinder- und Jugendbericht—Bericht über die Lebenssituation junger Menschen und die Leistungen der Kinder- und Jugendhilfe in Deutschland* [Twelfth Child and Youth Report—Report on the living conditions of young people and the benefits of child and youth services in Germany] (Bundestagsdrucksache, No. 15/6014). Berlin, Germany: Bundesministerium für Familie, Senioren, Frauen und Jugend [Federal Ministry for Family Affairs, Senior Citizens, Women and Youth].
- Brooks-Gunn, J., Han, W.-J., & Waldfogel, J. (2002). Maternal employment and child cognitive outcomes in the first three years of life: The NICHD study of early child care. *Child Development*, 73, 1052–1072.
- Buckles, K. S., & Hungeman, D. M. (2013). Season of birth and later outcomes: Old questions, new answers. *Review of Economics and Statistics*, 95, 711–724.
- Carta, F., & Rizzica, L. (2018). Early kindergarten, maternal labor supply and children's outcomes: Evidence from Italy. *Journal of Public Economics*, 158, 79–102.
- Cascio, E. U. (2015). *The promises and pitfalls of universal early education* (IZA World of Labor Report, No. 116). Retrieved from <https://doi.org/10.15185/izawol.116>
- Cascio, E. U. (2017). *Does universal preschool hit the target? Program access and preschool impacts* (NBER Working Paper No. 23215). Cambridge, MA: National Bureau of Economic Research.
- Cascio, E. U., & Schanzenbach, D. W. (2016). First in the class? Age and the education production function. *Education Finance and Policy*, 11, 225–250.
- Chartier, A. M., & Geneix, N. (2006). *Pedagogical approaches to early childhood education* (Paper commissioned for the EFA Global Monitoring Report 2007). Paris, France: UNESCO.
- Cornelissen, T., Dustmann, C., Raute, A., & Schönberg, U. (2018). Who benefits from universal child care? Estimating marginal returns to early child care attendance. *Journal of Political Economy*, 126, 2356–2409.
- Cunha, F., & Heckman, J. J. (2007). The technology of skill formation. *American Economic Review: Papers and Proceedings*, 97, 31–47.
- Cunha, F., Heckman, J. J., & Schennach, S. M. (2010). Estimating the technology of cognitive and noncognitive skill formation. *Econometrica*, 78, 883–931.
- Datta Gupta, N., & Simonsen, M. (2010). Non-cognitive child outcomes and universal high quality child care. *Journal of Public Economics*, 94, 30–43.
- Datta Gupta, N., & Simonsen, M. (2016). Academic performance and type of early childhood care. *Economics of Education Review*, 53, 217–229.
- Drange, N., & Havnes, T. (2019). Early childcare and cognitive development: Evidence from an assignment lottery. *Journal of Labor Economics*, 37, 581–620.
- Dustmann, C., Puhani, P. A., & Schönberg, U. (2017). The long-term effects of early track choice. *Economic Journal*, 127, 1348–1380.
- Elango, S., Garcia, J. L., Heckman, J. J., & Hojman, A. (2016). Early childhood education. In R. Moffitt (Ed.), *Means-tested transfer programs in the United States* (Vol. II, pp. 235–298). Chicago, IL: University of Chicago Press.

- Evers, A., Lewis, J., & Riedel, B. (2005). Developing child-care provision in England and Germany: Problems of governance. *Journal of European Social Policy*, 15, 195–209.
- Evers, A., & Sachße, C. (2002). Social care services for children and older people in Germany: Distinct and separate histories. In J. Baldock, A. Anttonen, & J. Sipilä (Eds.), *The young, the old and the state: Social care systems in five industrial nations* (pp. 55–80). Cheltenham, UK: Edward Elgar.
- Fan, E., Liu, J.-T., & Chen, Y.-C. (2017). Is the quarter of birth endogenous? New evidence from Taiwan, the US, and Indonesia. *Oxford Bulletin of Economics and Statistics*, 79, 1087–1124.
- Faust, G. (2006). Zum Stand der Einschulung und der neuen Schuleingangsstufe in Deutschland [The state of enrollment and the new school entrance level in Germany]. *Zeitschrift für Erziehungswissenschaft*, 9, 328–347.
- Felfe, C., & Lalive, R. (2018). Does early child care affect children's development? *Journal of Public Economics*, 159, 33–53.
- Felfe, C., & Zierow, L. (2018). From dawn till dusk: Implications of full-day care for children's development. *Labour Economics*, 55, 259–281.
- Frankfurter Allgemeine Zeitung (FAZ). (1996, June 25). Rechtsanspruch wird eingelöst [Legal claim is honored]. *Frankfurter Allgemeine Zeitung*, p. 38.
- Frankfurter Allgemeine Zeitung (FAZ). (1997, August 21). 600 Plätze sind ständig frei [600 slots are always free]. *Frankfurter Allgemeine Zeitung*, p. 37.
- Frankfurter Allgemeine Zeitung (FAZ). (1998, December 30). Genügend Kindergartenplätze [Enough kindergarten places]. *Frankfurter Allgemeine Zeitung*, p. 62.
- Garces, E., Thomas, D., & Currie, J. (2002). Longer-term effects of Head Start. *American Economic Review*, 92, 999–1012.
- Gormley, W. T., & Gayer, T. (2005). Promoting school readiness in Oklahoma an evaluation of Tulsa's pre-k program. *Journal of Human Resources*, 40, 533–558.
- Haecck, C., Lebihan, L., & Merrigan, P. (2018). Universal child care and long-term effects on child well-being: Evidence from Canada. *Journal of Human Capital*, 12, 38–98.
- Haecck, C., Lefebvre, P., & Merrigan, P. (2015). Canadian evidence on ten years of universal preschool policies: The good and the bad. *Labour Economics*, 36, 137–157.
- Hank, K., & Kreyenfeld, M. (2003). A multilevel analysis of child care and women's fertility decisions in Western Germany. *Journal of Marriage and Family*, 65, 584–596.
- Havnes, T., & Mogstad, M. (2011). No child left behind: Subsidized child care and children's long-run outcomes. *American Economic Journal: Economic Policy*, 3(2), 97–129.
- Havnes, T., & Mogstad, M. (2015). Is universal child care leveling the playing field? *Journal of Public Economics*, 127, 100–114.
- Heckman, J. J. (2008). Schools, skills, and synapses. *Economic Inquiry*, 46, 289–324.
- Heckman, J. J., Pinto, R., & Savelyev, P. (2013). Understanding the mechanisms through which an influential early childhood program boosted adult outcomes. *American Economic Review*, 103, 2052–2086.
- Heckman, J. J., Urzua, S., & Vytlačil, E. (2006). Understanding instrumental variables in models with essential heterogeneity. *Review of Economics and Statistics*, 88, 389–432.
- Heinze, R. G., Schmid, J., & Strünck, C. (1997). Zur politischen Ökonomie der sozialen Dienstleistungsproduktion. Der Wandel der Wohlfahrtsverbände und die Konjunkturen der Theoriebildung [The political economy of social service production: The transformation of charities and the cycles of theory formation]. *Kölner Zeitschrift für Soziologie und Sozialpsychologie*, 49, 242–271.
- Hofferth, S. L., & Wissoker, D. A. (1992). Price, quality, and income in child care choice. *Journal of Human Resources*, 27, 70–111.
- Hsin, A., & Felfe, C. (2014). When does time matter? Maternal employment, children's time with parents, and child development. *Demography*, 51, 1867–1894.
- Ichino, A., Fort, M., & Zanella, G. (Forthcoming). Cognitive and non-cognitive costs of daycare 0–2 for children in advantaged families. *Journal of Political Economy*.
- Imbens, G. W., & Rubin, D. B. (2015). *Causal inference in statistics, social, and biomedical sciences*. Cambridge, UK: Cambridge University Press.
- Kottelenberg, M. J., & Lehrer, S. F. (2014). Do the perils of universal childcare depend on the child's age? *CESifo Economic Studies*, 60, 338–365.
- Kottelenberg, M. J., & Lehrer, S. F. (2017). Targeted or universal coverage? Assessing heterogeneity in the effects of universal child care. *Journal of Labor Economics*, 35, 609–653.
- Kreyenfeld, M., & Hank, K. (2000). Does the availability of child care influence the employment of mothers? Findings from Western Germany. *Population Research and Policy Review*, 19, 317–337.
- Kreyenfeld, M., Spieß, C. K., & Wagner, G. (2001). *Finanzierungs- und Organisationsmodelle institutioneller Kinderbetreuung: Analysen zum Status quo und Vorschläge zur Reform* [Financing and organizational

- models of institutional childcare: Analyses of the status quo and proposals for reform]. Neuwied, Germany: Luchterhand.
- Lefebvre, P., & Merrigan, P. (2008). Child-care policy and the labor supply of mothers with young children: A natural experiment from Canada. *Journal of Labor Economics*, 26, 519–548.
- Lutz, W. (2014). A population policy rationale for the twenty-first century. *Population and Development Review*, 40, 527–544.
- Lutz, W., Butz, W. P., & KC, S. (2014). *World population and human capital in the twenty-first century*. Oxford, UK: Oxford University Press.
- Mamier, J., Pluto, L., van Santen, E., Seckinger, M., & Zink, G. (2002). Jugendhilfe und sozialer wandel [Youth welfare and social change]. *DJI Bulletin*, 59, 4–7.
- McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, 142, 698–714.
- Nollenberger, N., & Rodríguez-Planas, N. (2015). Full-time universal childcare in a context of low maternal employment: Quasi-experimental evidence from Spain. *Labour Economics*, 36, 124–136.
- Nores, M., & Barnett, W. S. (2010). Benefits of early childhood interventions across the world: (Under) Investing in the very young. *Economics of Education Review*, 29, 271–282.
- OECD. (2002). *Education at a glance 2002*. Paris, France: Organisation for Economic Co-Operation and Development.
- OECD. (2011). *Doing better for families*. Paris, France: Organisation for Economic Co-Operation and Development.
- OECD. (2016a). *OECD family database: Fertility rates*. Paris, France: OECD Social Policy Division, Directorate of Employment, Labour and Social Affairs.
- OECD. (2016b). *OECD family database: Maternal employment*. Paris, France: OECD Social Policy Division, Directorate of Employment, Labour and Social Affairs.
- OECD. (2017). *OECD family database: Enrolment in childcare and pre-school*. Paris, France: OECD Social Policy Division, Directorate of Employment, Labour and Social Affairs.
- Pei, Z., Pischke, J.-S., & Schwandt, H. (2019). Poorly measured confounders are more useful on the left than on the right. *Journal of Business and Economic Statistics*, 37, 205–216.
- Rege, M., Solli, I. F., Størksen, I., & Votruba, M. (2018). Variation in center quality in a universal publicly subsidized and regulated childcare system. *Labour Economics*, 55, 230–240.
- Rubin, D. B. (2001). Using propensity scores to help design observational studies: Application to the tobacco litigation. *Health Services & Outcomes Research Methodology*, 2, 169–188.
- Ruhm, C., & Waldfogel, J. (2012). Long-term effects of early childhood care and education. *Nordic Economic Policy Review*, 1(1), 23–51.
- Scheiwe, K., & Willekens, H. (2009). *Child care and preschool development in Europe: Institutional perspectives*. Basingstoke, UK: Palgrave Macmillan.
- Schneeweis, N., & Zweimüller, M. (2014). Early tracking and the misfortune of being young. *Scandinavian Journal of Economics*, 116, 394–428.
- Shonkoff, J. P., & Phillips, D. A. (Eds.). (2000). *From neurons to neighborhoods: The science of early childhood development*. Washington, DC: National Academies Press.
- Skoepel, J., Pink, S., & Bela, D. (2013). *Starting cohort 4: Grade 9 (SC4). SUF Version 1.1.0. data manual* (NEPS research data paper). Bamberg, Germany: University of Bamberg, National Educational Panel Study.
- Taylor, K. (2017, May 10). Is “3-K for all” good for all? De Blasio’s preschool plan troubles some. *The New York Times*. Retrieved from <https://www.nytimes.com/2017/05/10/nyregion/free-preschool-deblasio-new-york-city.html>
- Wagner, G. G., Frick, J. R., & Schupp, J. (2007). The German Socio-Economic Panel Study (SOEP)—Scope, evolution and enhancements. *Schmollers Jahrbuch: Journal of Applied Social Science Studies*, 127, 139–169.